Nondimensional Staged Feedback Control of Mitochondrial Heteroplasmy Dynamics with Host-Emulated Verification on an Ultra-Low-Power RV32IMF_Zve32x Neural Processing Unit

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Code & Reproducibility Repository: https://github.com/deesatzed/mitorestorer

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Abstract

Restoring bioenergetic function in cells burdened by mutant or damaged mitochondrial populations (D) alongside wild-type populations (W) requires navigating nonlinear couplings among autophagic clearance bottlenecks (B), reactive oxygen species (ROS) conversion, and replicative selection ($k_{\text{rep,adv}} \geq 1$). Here we formulate an eight-state ordinary differential equation (ODE) model with exact 1-to-1 correspondence between analytical equations, IEEE-754 `FP64` simulation, and bare-metal C99 firmware, analyzing a three-input staged feedback policy: Stage 0 bottleneck clearance (u_0), Stage 1 selective mitophagy (u_1), and Stage 2 biogenesis (u_2) gated jointly at $h(t) = D(t)/(W(t)+D(t)) \leq h_{\text{gate}} = 0.080$ and $B(t) \leq B_{\text{gate}} = 0.120$. Using Nagumo's subtangent condition, we prove joint forward invariance of the Stage-1 safe set ($\text{EP} \leq 1.0456 < \theta_{\text{GSH}} = 1.150$, $W_{\min} = 0.600 > \theta_{\text{surv}} = 0.450$, $\lambda_{\min} = 0.01085\ \text{d}^{-1}$) and derive both an exact accumulated-selection identity $s(t) = \int_0^t \lambda_h(q)\,dq$ (supremum error 0.0000, $s^*(0.40, 0.08) = 2.0369$) and a zero-tuning closed-form perturbation bound for constant-rate rescaling $\hat{\tau} = \hat{\lambda}_0 t$ across three regimes with distinct dimensionless Π-groups ($R^2 = 0.9828$, supremum error $0.0493 \leq 0.1216$). Solving the 300-variable discrete-adjoint optimal control problem (`L-BFGS-B`, gradient error 1.95×10^{-9}, 5-start spread 0.12%) shows that Pontryagin-optimal trajectories ride an interior mitophagy arc $u_1^*(t) \in (0, 1)$ along $\text{EP} = \theta_{\text{GSH}}$ before a terminal biogenesis sprint ($J^*_{\text{PMP}} = 0.7262$ vs. two-stage relay $J^*_{\text{2stage}} = 0.850$ and threshold feedback $J_{\text{thresh}} = 0.8875$), trading +22.21% open-loop Bolza cost for 20.1% redox margin. Mapping the 31×31 (W_0, k_{dmg}) phase plane delineates irreversible collapse (18.9%) and reconciles build-first rescue (2.9%) with purge-then-build control. Under four-channel proxy sensing ($\kappa(\mathcal{O}_7) = 745.6$), $\pm 20\%$ 7-rate mismatch, $\pm 8\%$ initial-state uncertainty, AR(1) noise, and 5% 6.5σ spikes, margin-tightened Huber-EKF ($\hat{h}_{\text{gate}} = 0.065$) reduces premature switches from 24.17% to 5.00%. On host-emulated Google Coral NPU (`RV32IMF_Zve32x`, `sizeof` static workspace = 3,696 B/8,192 B in L1 DTCM Bank 0), our C99 fixed-point Huber-EKF (0.94% $\hat{h}$ RMSE) and `Q16.16` `RK4` kernel with `Q8.24` slopes and $\Sigma\Delta$ remainder diffusion eliminate deadband stalling across a $16\times$ step sweep (0.0534%–0.0563% vs. 100% naive `Q16.16` stall at $N = 8{,}000$).

1 Introduction

1.1 Nonlinear Coupling in Mitochondrial Quality Control

Eukaryotic bioenergetics depends on maintaining a high functional fraction of intact wild-type mitochondrial DNA and cristae (W) relative to deletion-bearing or depolarized mutant organelles (D) [1, 2, 3]. Across diverse biological timescales—including post-mitotic dopaminergic neurons in Parkinsonian neurodegeneration ($T_{\max} = 90$ d) [4, 5], *ex vivo* chimeric antigen receptor (CAR) T-cell expansion bioreactors ($T_{\max} = 168$ h) [6, 7, 8], and honeybee (*Apis mellifera*) thoracic flight muscle exposed to strobilurin Complex III inhibition ($T_{\max} = 14$ d) [9, 10]—mitochondrial decline is governed by three interacting feedbacks:

1. **Upstream Clearance Bottlenecks (B):** Accumulation of α-synuclein oligomers, lipofuscin, or xenobiotic fungicides elevates energetic impedance and saturates autophagosome-lysosome flux $V_{\text{lys}}(B)$ [4, 11].

2. **Threshold ROS Conversion ($\text{EP} > \theta_{\text{GSH}}$):** When cellular energetic demand and impedance exceed the glutathione/peroxiredoxin antioxidant buffering threshold θ_{GSH}, electron leak generates reactive oxygen species (ROS) that convert intact wild-type mass W into damaged mass D.

3. **Asymmetric Replicative Selection ($k_{\text{rep,adv}}$):** Deletion-bearing mtDNA templates or fragmented

organelles frequently exhibit a replicative or transcription-coupled copy-number advantage ($k_{\text{rep,adv}} \geq 1$) during non-selective biogenesis [2, 12].

1.2 Comparison with Prior Control Formulations

In a foundational mathematical study, Hoitzing et al. [13] analyzed stochastic and deterministic mtDNA population control in a two-species (W, D) system without an upstream lysosomal bottleneck state $B(t)$ or replicative advantage ($k_{\text{rep,adv}} = 1$). When wild-type mass W_0 is already near the minimum survival floor and $B_0 \approx 0$, Hoitzing et al. demonstrated that an initial biogenesis pulse ("build-first") prevents transient ATP starvation before selective clearance. Conversely, when an upstream bottleneck $B_0 > 0$ elevates bioenergetic impedance and $k_{\text{rep,adv}} > 1$, stimulating biogenesis ($u_2 > 0$) prior to clearing B and D drives the energy potential $\text{EP}(t)$ above θ_{GSH} and expands the mutant pool. Understanding the precise mathematical boundary between these two regimes—and implementing a verifiable closed-loop controller within the strict memory and arithmetic constraints of an ultra-low-power implantable or bioreactor neural processing unit (NPU)—remains an open systems-and-control problem.

1.3 Contributions and Scope

This paper makes four self-contained, computationally verified contributions:

1. **Unified 8-State ODE Model and Joint Nagumo Safety Invariance (§2.1–§2.2):** We formulate an 8-state ODE plant with closed-form auxiliary rates unified 1-to-1 across LaTeX, Python, and C99, prove joint forward invariance of the Stage-1 safe box via Nagumo's subtangent condition (proving $\text{EP} \leq 1.0456 < \theta_{\text{GSH}}$ dynamically and $W_{\min} = 0.600 > \theta_{\text{surv}} = 0.450$, Lemma 1), and derive an analytical uniform contraction lower bound $\lambda_{\min} = 0.01085\ \text{d}^{-1}$ (Proposition 1).

2. **Multi-Start Adjoint Optimal Control and 2D Controllability Phase Diagram (§2.3–§2.4):** We solve the unconstrained 300-variable discrete-adjoint optimal control problem (`L-BFGS-B` across 5 diverse initializations, 0.12% cost spread) to decompose the optimality gap between Pontryagin interior-arc modulation ($J^*_{\text{PMP}} = 0.7262$), optimal two-stage constant relay control ($J^*_{\text{2stage}} = 0.850$), and closed-loop threshold feedback ($J_{\text{thresh}} = 0.8875$), and map the 31×31 (W_0, k_{dmg}) phase diagram delineating irreversible collapse (18.9%) and Hoitzing et al.'s build-first rescue wedge (2.9%).

3. **Accumulated-Selection Identity and Zero-Tuning Perturbation Bound (§3):** We separate the exact accumulated-selection coordinate $s(t) = \int_0^t \lambda_h(q)\, dq$ (zero reduction error in the sub-damage regime) from constant-rate rescaling $\hat{\tau} = \hat{\lambda}_0 t$ computed via a closed-form Stage-0/1 bottleneck integral across three regimes with distinct dimensionless Π-groups, verifying analytical error bounds in every regime (Proposition 2, Table 2).

4. **Mismatched Output-Feedback Huber-EKF and `RV32IMF_Zve32x` Firmware (§4–§5):** We evaluate the Stage-1 7-state Observability Gramian ($\kappa(\mathcal{O}_7) = 745.6$), benchmark both Python Monte Carlo and bare-metal C99 fixed-point Huber-EKF under $\pm 20\%$ kinetic mismatch and 6.5σ sensor spikes, prove a reversible Gaussian AR(1) Markov persistence bound (Proposition 3) with equal-footing plant-preserving actuator rollback ($\Delta h_{\max} = 0.0098$), and verify single-bank L1 DTCM residency (`sizeof` = 3,696 B) and deadband-free $\Sigma\Delta$ `RK4` integration (0.0534%–0.0563% error across $N \in [500, 8{,}000]$).

2 Mathematical Model and Staged Control Analysis

2.1 Eight-State Dynamical System and Closed-Form Auxiliary Rates

We model the intracellular mitochondrial network on a finite horizon $t \in [0, T_{\max}]$ via the state vector

$$\mathbf{x}(t) = \left[W,\, D,\, B,\, E,\, G,\, N,\, C,\, M_8\right]^\top \in \mathbb{R}^8_{\geq 0}, \qquad (1)$$

where $W(t)$ is normalized intact wild-type mitochondrial mass (healthy homeostatic baseline $W^* = 1.00$), $D(t)$ is normalized deletion-bearing or depolarized mutant mitochondrial mass, $B(t)$ is the upstream proteostatic/xenobiotic clearance bottleneck load, $E(t)$ is filtered cellular energy-resistance potential, $G(t)$ is a secreted/circulating metabolic stress readout (e.g., GDF15 or extracellular lactate accumulation), $N(t)$ is the normalized NAD^+/NADH redox cofactor pool, $C(t)$ is phosphocreatine/shuttle buffer capacity, and $M_8(t)$ is a downstream epigenetic/exhaustion drift readout state.

Let $M(t) = W(t) + D(t)$ denote total mitochondrial mass and $h(t) = D(t)/M(t) \in [0, 1)$ denote the mutant heteroplasmy fraction. The system is actuated by a bounded three-dimensional control vector $\mathbf{u}(t) = [u_0(t), u_1(t), u_2(t)]^\top \in [0, 1]^3$, where u_0 drives upstream bottleneck clearance, u_1 drives PINK1/Parkin-mediated selective mitophagy of D, and u_2 drives PGC-1α/TFAM-mediated mitochondrial biogenesis.

To guarantee exact 1-to-1 correspondence between the mathematical analysis, the floating-point benchmark suite (`eval_rhs_fp64`), and the bare-metal C99 fixed-point kernel (`coralnpu_eval_core8_rhs_q16`), all auxiliary rates are defined in closed form without unmodeled

factors:

$$V_{\mathrm{lys}}(B) = \frac{K_B}{K_B + B}, \tag{2}$$

$$k_{\mathrm{bio}}(u_2, N) = k_{\mathrm{bio},0} + (k_{\mathrm{bio,max}} - k_{\mathrm{bio},0})\, u_2\, \frac{N}{1+N}, \tag{3}$$

$$k_{\mathrm{mito},D}(u_1) = k_{\mathrm{mito},W} + (k_{\mathrm{mito},D,\max} - k_{\mathrm{mito},W})\, u_1, \tag{4}$$

$$\mathrm{ATP}(\mathbf{x}) = (W + \eta_D D)(1 + \gamma_C C), \tag{5}$$

$$\dot{e}_R(\mathbf{x}) = \frac{1 + \beta_B B + \beta_h h}{\max(0.10,\, W + \eta_D D)(1 + \gamma_C C)}, \tag{6}$$

$$f_{\mathrm{eff}}(u_1, u_2) = 1 - c_1 u_1 + c_2 u_2, \tag{7}$$

$$\mathrm{EP}(\mathbf{x}, \mathbf{u}) = \dot{e}_R(\mathbf{x})\, f_{\mathrm{eff}}(u_1, u_2)^2. \tag{8}$$

Here $V_{\mathrm{lys}}(B) \in (0, 1]$ is the Michaelis-Menten lysosomal competence factor with half-saturation $K_B > 0$; $k_{\mathrm{bio},0}$ is the unstimulated basal biogenesis rate calibrated so that $(W^*, D^*) = (1.00, 0)$ is the exact basal equilibrium under $k_{\mathrm{mito},W}$:

$$k_{\mathrm{bio},0}\left(1 - \frac{W^*}{K}\right) = k_{\mathrm{mito},W} \implies k_{\mathrm{bio},0} = \frac{k_{\mathrm{mito},W}}{1 - 1/K}. \tag{9}$$

For $K = 1.250$ and $k_{\mathrm{mito},W} = 0.015\ \mathrm{d}^{-1}$, Eq. (9) yields the exact basal turnover rate $k_{\mathrm{bio},0} = 0.075\ \mathrm{d}^{-1}$.

The eight-state ODE system is given by:

$$\dot{W} = k_{\mathrm{bio}}(u_2, N)\, W \left(1 - \frac{W + D}{K}\right) - \Big(k_{\mathrm{mito},W} + k_{\mathrm{dmg}} \max\big(0, \mathrm{EP} - \theta_{\mathrm{GSH}}\big)\Big) W, \tag{10}$$

$$\dot{D} = k_{\mathrm{rep,adv}}\, k_{\mathrm{bio}}(u_2, N)\, D \left(1 - \frac{W + D}{K}\right) - k_{\mathrm{mito},D}(u_1)\, V_{\mathrm{lys}}(B)\, D + k_{\mathrm{dmg}} \max\big(0, \mathrm{EP} - \theta_{\mathrm{GSH}}\big) W, \tag{11}$$

$$\dot{B} = b_{\mathrm{basal}} + k_{\mathrm{acc}}\, D\, (1 - u_0) - k_{\mathrm{clr}}\, u_0\, B, \tag{12}$$

$$\dot{E} = k_E\big(\mathrm{EP}(\mathbf{x}, \mathbf{u}) - E\big), \tag{13}$$

$$\dot{G} = k_{G,\mathrm{prod}}\, E - k_{G,\mathrm{clr}}\, G, \tag{14}$$

$$\dot{N} = k_{N,\mathrm{syn},0} + k_{N,\mathrm{syn},2}\, u_2 - k_{\mathrm{PARP}} \max\big(0, E - \theta_{\mathrm{GSH}}\big) N - k_{N,\mathrm{use}}\, N, \tag{15}$$

$$\dot{C} = (k_{\mathrm{oct},0} + k_{\mathrm{oct},2}\, u_2)(1.30 - C) - k_{\mathrm{ox}} \max\big(0, E - \theta_{\mathrm{cys}}\big) C, \tag{16}$$

$$\dot{M}_8 = k_{\mathrm{drift}}\, E\, (10.0 - M_8) - k_{\mathrm{TET}}\, u_2\, N\, C\, M_8. \tag{17}$$

Observation Model and Structural Observability Collinearity. Because intracellular masses (W, D) and bottleneck burden B are not directly measurable at real-time sampling rates on an embedded processor, the controller observes a discrete-time four-channel proxy vector $\mathbf{y}_k \in \mathbb{R}^4$ at sampling instants t_k:

$$\mathbf{y}_k = \mathbf{h}_{\mathrm{obs}}\big(\mathbf{x}(t_k)\big) + \mathbf{v}_k = \begin{bmatrix} E(t_k) \\ \mathrm{ATP}\big(\mathbf{x}(t_k)\big) \\ G(t_k) \\ N(t_k) \end{bmatrix} + \mathbf{v}_k, \tag{18}$$

where $y_{k,1}$ tracks filtered energetic impedance/stress potential, $y_{k,2}$ is a respirometric or metabolic ATP proxy, $y_{k,3}$ is secreted stress readout, and $y_{k,4}$ is optical NAD(P)H autofluorescence. Crucially, Eqs. (5)–(8) imply $\mathrm{EP} \cdot \mathrm{ATP}/f_{\mathrm{eff}}^2 = 1 + \beta_B B + \beta_h h$: at any single snapshot t_k, B and h enter the static observation map only as a linear combination, while (W, D) and C exhibit multiplicative iso-ATP trade-offs in Eq. (5) (and M_8 is a purely downstream readout with zero feedback into states $x_0 \dots x_6$). Reconstructing $(\hat{W}, \hat{D}, \hat{B})$ therefore relies on multi-step dynamic separation via a 7-state Huber-regularized extended Kalman filter (Huber-EKF) [14, 15], whose local Observability Gramian conditioning and sensitivity to $\pm 20\%$ plant-model mismatch we quantify explicitly in §5.2.

2.2 Exact Quotient Flow and Conditional Safety Guarantees

Applying the quotient rule $\frac{d}{dt}\left(\frac{D}{W+D}\right) = \frac{\dot{D}W - D\dot{W}}{(W+D)^2}$ to Eqs. (10)–(11) yields the exact heteroplasmy quotient ODE. Specifically, the ROS conversion term $+k_{\mathrm{dmg}} W \max(0, \mathrm{EP} - \theta_{\mathrm{GSH}})$ in $\dot{D}$ contributes $\frac{W^2}{(W+D)^2} k_{\mathrm{dmg}} \max(0, \mathrm{EP} - \theta_{\mathrm{GSH}}) = (1-h)^2 k_{\mathrm{dmg}} \max(0, \mathrm{EP} - \theta_{\mathrm{GSH}})$, while the corresponding loss term $-k_{\mathrm{dmg}} W \max(0, \mathrm{EP} - \theta_{\mathrm{GSH}})$ in $\dot{W}$ contributes $+\frac{DW}{(W+D)^2} k_{\mathrm{dmg}} \max(0, \mathrm{EP} - \theta_{\mathrm{GSH}}) = h(1-h) k_{\mathrm{dmg}} \max(0, \mathrm{EP} - \theta_{\mathrm{GSH}})$. Summing both contributions yields $(1-h) k_{\mathrm{dmg}} \max(0, \mathrm{EP} - \theta_{\mathrm{GSH}})$, giving:

$$\dot{h} = -\lambda_h(t)\, h(1-h) + (1-h)\, k_{\mathrm{dmg}}\, \max\big(0, \mathrm{EP}(\mathbf{x}, \mathbf{u}) - \theta_{\mathrm{GSH}}\big), \tag{19}$$

$$\lambda_h(t) = k_{\mathrm{mito},D}(u_1)\, V_{\mathrm{lys}}(B) - k_{\mathrm{mito},W} - (k_{\mathrm{rep,adv}} - 1)\, k_{\mathrm{bio}}(u_2, N) \left(1 - \frac{M(t)}{K}\right). \tag{20}$$

Equation (20) reveals why concurrent or premature biogenesis ($u_2 = 1$) can fail when $k_{\mathrm{rep,adv}} > 1$ and $B(t)$ is elevated: increasing $k_{\mathrm{bio}}(u_2, N)$ directly subtracts $(k_{\mathrm{rep,adv}} - 1) k_{\mathrm{bio}}(1 - M/K)$ from the selective contraction rate $\lambda_h(t)$, while elevated $B(t)$ depresses $V_{\mathrm{lys}}(B)$ and raises $\mathrm{EP}(\mathbf{x}, \mathbf{u})$ above θ_{GSH}.

We now prove joint forward invariance of the Stage-1 safe box via Nagumo's subtangent theorem, dynamically establishing $\mathrm{EP}(\mathbf{x}, (1, 1, 0)) < \theta_{\mathrm{GSH}}$ rather than assuming it *a priori*, and resolving the exact boundary vs. interior dependence on B_0.

Lemma 1 (Joint Stage-1 Safe-Box Invariance) *Under Stage 1 control ($u_0 = 1, u_1 = 1, u_2 = 0$), Eqs. (3)–(4) give identically $k_{\mathrm{bio}}(0, N) = k_{\mathrm{bio},0}$ and $k_{\mathrm{mito},D}(1) = k_{\mathrm{mito},D,\max}$. For $D_0 \in (0, W^*)$ and $B_0 \geq B_{\mathrm{eq}} = b_{\mathrm{basal}}/k_{\mathrm{clr}}$, define the invariant wild-type floor $W_{\min}(D_0) \triangleq K(1 - k_{\mathrm{mito},W}/k_{\mathrm{bio},0}) - D_0 = W^* - D_0$, the candidate compact box $\Omega_1(D_0, B_0) \triangleq \{\mathbf{x} \in \mathbb{R}^8_{\geq 0}$:*

Table 1: Exact Model Parameters, Physical Units, and Provenance Classification for the Canonical Neuronal Regime ($T_{\max} = 90$ d, matching `DEFAULT_NEURON_PARAMS` in Python and `p_q16` in C99). Classification: **M** (Measured order-of-magnitude in literature), **F** (Fitted/calibrated to equilibrium or turnover constraints), **A** (Control design specification).

Symbol	Description	Value	Units	Class	Literature Basis / Calibration Rationale
K	Mitochondrial carrying capacity	1.250	Norm. mass	**F**	25% reserve above homeostatic $W^* = 1.00$ [2]
$k_{\mathrm{mito},W}$	Basal wild-type mitophagy rate	0.015	d^{-1}	**M**	Neuronal mtDNA half-life $t_{1/2} \approx 46$ d [3, 4]
$k_{\mathrm{bio},0}, k_{\mathrm{bio,max}}$	Basal / max stimulated biogenesis rates	$0.075, 0.140$	d^{-1}	**F**	Exact basal equilibrium $k_{\mathrm{mito},W}/(1-1/K)$ (Eq. 9) [11]
$k_{\mathrm{mito},D,\max}$	Max Stage-1 selective mitophagy rate	0.115	d^{-1}	**M**	Stimulated PINK1/Parkin clearance ($t_{1/2} \approx 6$ d) [11]
$k_{\mathrm{rep,adv}}$	Mutant replicative/survival advantage	1.320	Dimensionless	**M**	Deletion mtDNA expansion advantage (swept 1.00–1.32) [2, 12]
k_{dmg}	Supra-threshold ROS conversion rate	0.040	$\mathrm{d}^{-1}\mathrm{EP}^{-1}$	**F**	Progressive untreated ROS conversion (swept 0.00–0.045 in Fig. 2B) [4]
$\theta_{\mathrm{GSH}}, \theta_{\mathrm{cys}}$	Antioxidant & thiol oxidation thresholds	$1.150, 0.850$	Norm. EP	**F**	Glutathione/peroxiredoxin redox ceiling above $\mathrm{EP}^* \le 1.00$ [5]
θ_{surv}	Minimum bioenergetic survival floor	0.450	Norm. ATP	**M**	Apoptotic/necrotic ATP threshold (45% of baseline) [5]
K_B	Lysosomal bottleneck half-saturation	0.250	Norm. load	**F**	Michaelis-Menten autophagic flux saturation [4]
$b_{\mathrm{basal}}, k_{\mathrm{acc}}, k_{\mathrm{clr}}$	Bottleneck basal influx, D-coupling, clearance	$0.0072, 0.080, 0.180$	d^{-1}	**F**	$b_{\mathrm{basal}} = 472/2^{16}$; cleared floor $B_{\mathrm{eq}} = b_{\mathrm{basal}}/k_{\mathrm{clr}} = 0.040$
$\beta_B, \beta_h, \eta_D, \gamma_C$	Impedance & OXPHOS coupling coefficients	$0.25, 0.65, 0.15, 0.34$	Dimensionless	**F**	Defective OXPHOS coupling of deletion-bearing D [1]
c_1, c_2	Mitophagy / biogenesis flux load factors	$0.220, 0.040$	Dimensionless	**F**	Energetic relief during mitophagy vs. biogenesis translation load
$k_E, k_{G,\mathrm{prod}}, k_{G,\mathrm{clr}}$	Potential relaxation & stress readout rates	$0.250, 0.080, 0.160$	d^{-1}	**F**	First-order cellular potential and biomarker kinetics
$k_{N,\mathrm{syn},0/2}, k_{\mathrm{PARP}}, k_{N,\mathrm{use}}$	NAD^+ synthesis, PARP1 drain, basal use	$0.060, 0.080, 0.060, 0.095$	d^{-1}	**F**	PARP1-mediated NAD^+ consumption above θ_{GSH} [11]
$k_{\mathrm{oct},0/2}, k_{\mathrm{ox}}, k_{\mathrm{drift}}, k_{\mathrm{TET}}$	Buffer (C) & epigenetic (M_8) kinetics	$0.04, 0.10, 0.05, 0.0015, 0.012$	d^{-1}	**F**	Thiol oxidation and cofactor-dependent demethylation kinetics
$h_{\mathrm{gate}}, B_{\mathrm{gate}}$	Stage-1 → Stage-2 switching thresholds	$0.080, 0.120$	Dimensionless	**A**	Derived from Eq. (29); tightened to 0.065 under mismatch (§5.2)

$W \in [W^* - D_0, K]$, $D \in [0, D_0]$, $B \in [B_{\mathrm{eq}}, B_0]$, $E \in [0, E_{\max}]$, $C \in [C_{\min}, 1.30]\}$, *and the boundary vs. interior critical bottleneck thresholds:*

$$B^*_{\mathrm{crit}} \triangleq K_B \left(\frac{k_{\mathrm{mito},D,\max}}{k_{\mathrm{rep,adv}}\, k_{\mathrm{bio},0}\,(1 - W^*/K)} - 1 \right), \tag{21}$$

$$B_{\mathrm{crit,mono}} \triangleq K_B \left(\frac{k_{\mathrm{mito},D,\max}}{k_{\mathrm{rep,adv}}\, k_{\mathrm{bio},0}\,\left(1 - \frac{W^* - D_0}{K}\right)} - 1 \right). \tag{22}$$

Suppose $\sup_{\Omega_1} \mathrm{EP}(\mathbf{x}, (1,1,0)) \le \min(\theta_{\mathrm{GSH}}, E_{\max})$ *and* $k_{\mathrm{oct},0}(1.30 - C_{\min}) \ge k_{\mathrm{ox}} \max(0, E_{\max} - \theta_{\mathrm{cys}}) C_{\min}$. *Then:*

1. ***Joint Forward Invariance*** *($B_0 \le B^*_{\mathrm{crit}}$): The compact set $\Omega_1(D_0, B_0)$ is positively invariant under Eqs. (10)–(16). In particular, for all $t \in [0, t_{\mathrm{gate}}]$, $\mathrm{EP}(\mathbf{x}(t), (1,1,0)) \le \theta_{\mathrm{GSH}}$, $D(t) \le D_0$, and $W(t)$ satisfies*

$$W(t) \ge W_{\min}(D_0) = W^* - D_0. \tag{23}$$

Moreover, once $B(t) \le B_{\mathrm{crit,mono}}$, $\dot{D}(t) \le 0$ holds pointwise throughout the entire interior of $\Omega_1(D_0, B_0)$.

2. ***Super-Critical Bottleneck Dwell Time*** *($B_0 > B^*_{\mathrm{crit}}$): Since $\dot{B} = b_{\mathrm{basal}} - k_{\mathrm{clr}} B$ under $u_0 = 1$, $B(t)$ reaches B^*_{crit} at the finite dwell time*

$$t_0 = \frac{1}{k_{\mathrm{clr}}} \ln\left(\frac{B_0 - B_{\mathrm{eq}}}{B^*_{\mathrm{crit}} - B_{\mathrm{eq}}} \right), \tag{24}$$

with $D(t_0) \le \bar{D}(t_0) \triangleq \min\big(K, D_0 \exp(k_{\mathrm{rep,adv}} k_{\mathrm{bio},0} t_0)\big)$, after which Case 1 invariance holds on $\Omega_1(\bar{D}(t_0), B^*_{\mathrm{crit}})$ whenever $\bar{D}(t_0) < W^*$ and $\mathrm{EP} \le \theta_{\mathrm{GSH}}$.

Proof. We verify Nagumo's subtangent condition on every boundary face of $\Omega_1(D_0, B_0)$. First, by assumption $\mathrm{EP}(\mathbf{x}, (1,1,0)) \le \min(\theta_{\mathrm{GSH}}, E_{\max})$ everywhere on Ω_1, so the ROS damage term $\max(0, \mathrm{EP} - \theta_{\mathrm{GSH}}) \equiv 0$ on Ω_1. On the E, C, and B faces: $\dot{E}|_{E=0} = k_E \mathrm{EP} \ge 0$, $\dot{E}|_{E=E_{\max}} = k_E(\mathrm{EP} - E_{\max}) \le 0$; $\dot{C}|_{C=1.30} \le 0$, $\dot{C}|_{C=C_{\min}} \ge 0$; and under $u_0 = 1$, $\dot{B}|_{B=B_{\mathrm{eq}}} = 0$ and $\dot{B}|_{B=B_0} = k_{\mathrm{clr}}(B_{\mathrm{eq}} - B_0) \le 0$. Crucially, on the upper mutant face $D = D_0$, every point in Ω_1 satisfies $W \ge W^* - D_0$, so total mass on that face satisfies $M = W + D_0 \ge W^*$ automatically (without assuming $M(t) \ge W^*$ in the interior). Evaluating Eq. (11) at $D = D_0$ therefore gives $\dot{D}|_{D=D_0} \le D_0 \big[k_{\mathrm{rep,adv}} k_{\mathrm{bio},0}(1 - W^*/K) - k_{\mathrm{mito},D,\max} \frac{K_B}{K_B + B}\big] \le 0$ for all $B \le B^*_{\mathrm{crit}}$, while $\dot{D}|_{D=0} = 0$. On the lower wild-type face $W = W^* - D_0$, every point in Ω_1 satisfies $D \le D_0$, so Eq. (10) gives

$$\dot{W}\big|_{W_{\min}} \ge W_{\min} \left[k_{\mathrm{bio},0} \left(1 - \frac{W^*}{K} \right) - k_{\mathrm{mito},W} \right] = 0 \tag{25}$$

by Eq. (9), and $\dot{W}|_{W=K} \le -k_{\mathrm{mito},W} K < 0$. Thus $\Omega_1(D_0, B_0)$ is jointly forward invariant. Throughout the interior of Ω_1, $M = W + D \ge W^* - D_0$, so $B(t) \le B_{\mathrm{crit,mono}}$ implies $\dot{D}(t) \le 0$ everywhere on Ω_1. Integrating $\dot{B} = b_{\mathrm{basal}} - k_{\mathrm{clr}} B$ yields Eq. (24). ∎

Exact Verification on the Canonical Neuronal Box. Set Ω_1 with $D_0 = 0.400$, $W \in [0.600, 1.250]$, $B \in [0.04001, 0.500]$, $E_{\max} = 1.350$, and $C_{\min} = 0.650$. Under $(u_0, u_1, u_2) = (1,1,0)$, $f_{\mathrm{eff}}(1,0)^2 = (1 - 0.220)^2 = 0.6084$. In Eq. (6), EP decreases strictly with W and C and increases with B; fixing $(W, B, C) = (0.600, 0.500, 0.650)$, $\phi(D) = \frac{1.125 + 0.65D/(0.60+D)}{0.60 + 0.15D} = \frac{0.675 + 1.775D}{0.36 + 0.69D + 0.15D^2}$ has numerator derivative $0.17325 - 0.2025D - 0.26625D^2 > 0$ for all $D \in [0, 0.400]$. Hence EP attains its exact global maximum on Ω_1 at $(0.600, 0.400, 0.500, 0.650)$:

$$\begin{aligned} \max_{\Omega_1} \mathrm{EP}\big(\mathbf{x}, (1,1,0)\big) &= \frac{1.385}{0.660 \times 1.221} \times 0.6084 \\ &= 1.0456 < \theta_{\mathrm{GSH}} = 1.150, \end{aligned}$$

while $\dot{C}|_{C=0.650} \ge 0.040(0.650) - 0.050(0.500)(0.650) = +0.00975 > 0$. Equations (21)–(22) give $B^*_{\mathrm{crit}} = 0.250(5.8081 - 1) = 1.2020 > B_0 = 0.500$ and $B_{\mathrm{crit,mono}} = 0.250(2.2340 - 1) = 0.3085$ (crossed by $B(t)$ at $t = 3.16$ d), confirming $W(t) \ge W_{\min}(0.40) = 0.600 > \theta_{\mathrm{surv}} = 0.450$

for all $t \in [0, t_{\text{gate}}]$. For super-critical $B_0 = 2.00 > 1.2020$, Eq. (24) gives $t_0 = 2.91$ d.

Proposition 1 (Analytical Regional Contraction) *Let $\mathbf{x}(0) \in \Omega_1(D_0, B_0)$ satisfy the hypotheses of Lemma 1 Case 1 ($B_0 \le B^*_{\text{crit}}$).*

1. ***Closed-Form Uniform Contraction Floor $\lambda_{\min}$:*** *On the invariant set $\Omega_1(D_0, B_0)$, $B(t) \le B_0$ and $M(t) = W(t) + D(t) \ge W^* - D_0$, so Eq. (20) under Stage 1 ($u_1 = 1, u_2 = 0$) is bounded below analytically for all $t \in [0, t_{\text{gate}}]$ by*

$$\lambda_{\min} \triangleq \frac{k_{\text{mito},D,\max} K_B}{K_B + B_0} - k_{\text{mito},W} - (k_{\text{rep,adv}} - 1) k_{\text{bio},0} \left(1 - \frac{W^* - D_0}{K}\right). \quad (26)$$

Whenever $\lambda_{\min} > 0$ (for the canonical neuronal box, $\lambda_{\min} = \frac{0.115}{3} - 0.015 - 0.32(0.075)(0.520) = 0.01085\ d^{-1} > 0$), $V(h) = \frac{1}{2}h^2$ satisfies $\dot{V} = -\lambda_h(t) h^2(1-h) \le -\lambda_{\min}(1-h_0)h^2$, yielding both the logistic and regional exponential bounds:

$$h(t) \le \frac{h_0\, e^{-\lambda_{\min} t}}{1 - h_0 + h_0\, e^{-\lambda_{\min} t}} \le h_0 \exp\big(-\lambda_{\min}(1-h_0)t\big). \quad (27)$$

2. ***Sub-Damage Transcritical Invasion Exchange:*** *At the mutant-free manifold $(W^*, 0, B_{\text{eq}})$ where $1 - W^*/K = k_{\text{mito},W}/k_{\text{bio}}$, the linearized mutant invasion eigenvalue is $\mu_D = -\lambda^*_h = k_{\text{rep,adv}}\, k_{\text{mito},W} - k_{\text{mito},D}(u_1)\, V_{\text{lys}}(B_{\text{eq}})$. Whenever $k_{\text{mito},D}(u_1)\, V_{\text{lys}}(B_{\text{eq}}) < k_{\text{rep,adv}}\, k_{\text{mito},W}$, $h^* = 0$ is unstable ($\mu_D > 0$); Stage 1 mitophagy satisfying $k_{\text{mito},D}(u_1)\, V_{\text{lys}}(B_{\text{eq}}) > k_{\text{rep,adv}}\, k_{\text{mito},W}$ renders $\mu_D < 0$ and $h^* = 0$ locally exponentially stable.*

Proof. By Lemma 1, $\mathbf{x}(t) \in \Omega_1(D_0, B_0)$ and $\text{EP}(t) \le \theta_{\text{GSH}}$ for all $t \in [0, t_{\text{gate}}]$, so Eq. (19) reduces to $\dot{h} = -\lambda_h(t) h(1-h)$, and substituting $V_{\text{lys}}(B(t)) \ge \frac{K_B}{K_B + B_0}$ and $1 - M(t)/K \le 1 - (W^* - D_0)/K$ into Eq. (20) proves Eq. (26). Separating variables gives $\frac{d}{dt} \ln\left(\frac{h}{1-h}\right) = -\lambda_h(t) \le -\lambda_{\min}$, which integrates to the first inequality in Eq. (27); since $h(t) \le h_0$ on $[0, t_{\text{gate}}]$, $1 - h(t) \ge 1 - h_0$, giving $\dot{h} \le -\lambda_{\min}(1-h_0)h$ and the second inequality. Linearizing $\dot{D}$ around $(W^*, 0, B_{\text{eq}})$ gives $\dot{D} \approx \mu_D D$. ■

2.3 Unconstrained Discrete-Adjoint Optimal Control vs. Threshold Feedback

To evaluate control policies over $t \in [0, T_{\max}]$ identically to `compute_bolza_cost` in `simulator/generate_paper1_verified_benchmarks.py`, we define the finite-horizon Bolza objective functional:

$$J(\mathbf{u}) = w_h\, h(T_{\max})^2 + w_W \big(W_{\text{ref}} - W(T_{\max})\big)^2 + \int_0^{T_{\max}} \Big[w_{\text{EP}}\, \max\big(0, \text{EP}(t) - \theta_{\text{GSH}}\big)^2 + w_{\text{surv}}\, \max\big(0, \theta_{\text{surv}} - \text{ATP}(t)\big)^2 + \frac{w_{\text{def}}}{T_{\max}}\, \max\big(0, 0.85 - W(t)\big)^2 + r_1 u_1(t) + r_2 u_2(t) \Big] dt, \quad (28)$$

with terminal weights $w_h = 80, w_W = 30$ ($W_{\text{ref}} = 1.00$), safety penalty weights $w_{\text{EP}} = 180\ \text{d}^{-1}, w_{\text{surv}} = 250\ \text{d}^{-1}, w_{\text{def}} = 0.85$, and horizon-normalized actuation weights $r_1 = 1.15/T_{\max}, r_2 = 0.12/T_{\max}$ (with $r_0 = 0$, representing non-penalized housekeeping bottleneck clearance so $\partial \mathcal{H}/\partial u_0 \le 0 \implies u_0^*(t) \equiv 1$).

Discrete-Adjoint Pontryagin Solution and Interior Constraint Arc. In Eqs. (10)–(17), (u_1, u_2) enter linearly in the actuation penalty $r_1 u_1 + r_2 u_2$ and in Eqs. (3)–(4), but enter *quadratically* in $\text{EP}(\mathbf{x}, \mathbf{u}) = \dot{e}_R(\mathbf{x})(1 - c_1 u_1 + c_2 u_2)^2$ whenever EP approaches $\theta_{\text{GSH}} = 1.150$, while u_2 also couples dynamically through cofactor $N(t)$ in Eq. (15). Monotonicity of $h(t)$ therefore neither establishes bang-bang optimality nor excludes interior state-constraint arcs.

To resolve the true optimal trajectory without structural assumptions, we discretize $\mathbf{u}(t) \in [0,1]^3$ into $N_{\text{ctrl}} = 100$ free piecewise-constant control intervals (300 decision variables over 1,000 exact `RK4` intervals on $[0, 90]$ d]), propagate the exact backward-in-time discrete `RK4` adjoint equations to obtain $\nabla_u J$ (verified against central finite differences to directional relative error 1.95×10^{-9}), and solve $\min_{u \in [0,1]^{300}} J(\mathbf{u})$ via `L-BFGS-B`. Across five diverse initializations (staged relay, all-zero, uniform midpoint 0.5, build-first, and uniform random), `L-BFGS-B` converges to the same optimum within a 0.12% cost spread ($J \in [0.72620, 0.72709]$, interior-arc stationarity residual $\max|\partial J/\partial u_{1,m}| = 1.47 \times 10^{-3}$). As shown in Figure 2A, the unconstrained optimum achieves $J^*_{\text{PMP}} = 0.7262$ ($h(T_{\max}) = 0.0300, W(T_{\max}) = 0.9938, \text{EP}_{\max} = 1.150$) via two non-relay features:

1. **Early Interior Redox-Boundary Modulation ($t \in [0, 17$ d$]$):** Rather than saturating $u_1 = 1$, $u_1^*(t)$ modulates smoothly across $[0.44, 0.83]$ so that $\text{EP}(t)$ rides just along $\theta_{\text{GSH}} = 1.150$ without incurring unnecessary $r_1 u_1$ effort, ramping to $u_1^* \approx 1.0$ during peak selection ($t \in [20, 33$ d$]$) before tapering to zero by $t \approx 75$ d.

2. **Late Terminal Biogenesis Sprint ($t \in [78, 90$ d$]$):** $u_2^*(t)$ remains identically zero until $h^*(t) < 0.030$, then executes a concentrated terminal burst to reach $W(T_{\max}) = 0.994$ while minimizing replicative expansion $(k_{\text{rep,adv}} - 1) k_{\text{bio}} D$.

Embedded State-Feedback Threshold Law and Optimality Decomposition. While $u^*(t)$ minimizes Eq. (28) in open loop, riding $\mathrm{EP}^*(t) \equiv \theta_{\mathrm{GSH}} = 1.150$ leaves zero robustness margin against kinetic mismatch or mid-horizon insults. For embedded closed-loop control, we trigger the Stage 1 $\to$ Stage 2 relay transition ($(u_0, u_1, u_2) : (1, 1, 0) \to (1, u_{1,\mathrm{S2}}, 1)$ with $u_{1,\mathrm{S2}} = 0.35$) when $B(t) \le B_{\mathrm{gate}} = 0.120$ and $h(t) \le h_{\mathrm{gate}}$ (`coralnpu_core8_stage_gate_q16`). By Eqs. (6)–(8), switching raises f_{eff}^2 from $(1 - c_1)^2 = 0.6084$ to $(1 - c_1 u_{1,\mathrm{S2}} + c_2)^2 = 0.9274$, requiring:

$$\mathrm{EP}(t_{\mathrm{gate}}^+) = \frac{1 + \beta_B B(t_{\mathrm{gate}}) + \beta_h h_{\mathrm{gate}}}{W(t_{\mathrm{gate}})\left(1 + \eta_D \frac{h_{\mathrm{gate}}}{1 - h_{\mathrm{gate}}}\right)\left(1 + \gamma_C C(t_{\mathrm{gate}})\right)} \times (1 - c_1 u_{1,\mathrm{S2}} + c_2)^2 \le \theta_{\mathrm{GSH}}. \quad (29)$$

At $h_{\mathrm{gate}} = 0.080$ ($t_{\mathrm{gate}} = 33.30$ d, $W = 0.772, B = 0.041, C = 1.091$), $\mathrm{EP}(t_{\mathrm{gate}}^+)$ jumps from 0.603 to $0.919 < \theta_{\mathrm{GSH}} = 1.150$, preserving a 20.1% redox safety margin (and remaining $\le 1.031 < 1.150$ even at unrecovered $C_0 = 0.65$). Conversely, for $h_{\mathrm{gate}} \ge 0.195$, $\mathrm{EP}(t_{\mathrm{gate}}^+)$ breaches $\theta_{\mathrm{GSH}} = 1.150$ and causes $J(h_{\mathrm{gate}})$ to spike sharply ($0.887 \to 3.120$, Figure 1B). Comparing $J_{\mathrm{PMP}}^* = 0.7262$, the optimal two-stage constant relay $J_{2\mathrm{stage}}^* = 0.850$ ($t_{\mathrm{sw}}^* = 48.18$ d, $u_{1,\mathrm{S2}}^* = 0$), and $J_{\mathrm{thresh}}(0.08) = 0.8875$ decomposes the suboptimality into a +17.05% structural relay-restriction cost and a +4.38% constant-threshold penalty (+22.21% total vs. J_{PMP}^*). Scaling the mitophagy weight $r_1 \leftarrow \alpha_{r_1} r_1$ across $\alpha_{r_1} \in \{0.25, 1.00, 4.00\}$ yields $(J_{\mathrm{PMP}}^*, J_{\mathrm{thresh}}) = (0.2231, 0.3782)$, $(0.7262, 0.8875)$, and $(2.4121, 2.9247)$ (+69.46%, +22.21%, +21.25% relative gaps), showing that the relative gap widens at very small r_1 due to fixed Stage-2 maintenance $u_{1,\mathrm{S2}} = 0.35$.

2.4 Competing Policies, Hoitzing Reconciliation, and 2D Collapse Phase Diagram

Figure 1A–B and Table 3A compare six control strategies from the canonical neuronal state ($W_0 = 0.60, D_0 = 0.40, B_0 = 0.50$): Untreated ($\mathrm{EP}_{\max} = 20.605$, $W(T_{\max}) = 0, J = 2{,}768{,}855.31$), Concurrent ($\mathrm{EP}_{\max} = 1.423 > \theta_{\mathrm{GSH}} = 1.150$, $J = 99.829$), Build-First at $B_0 = 0.50$ ($\mathrm{EP}_{\max} = 9.744$, $W(T_{\max}) = 0, J = 760{,}388.40$), Staged Threshold ($h_{\mathrm{gate}} = 0.080$: $\mathrm{EP}_{\max} = 1.046 < 1.150$, $W(T_{\max}) = 1.0487, h(T_{\max}) = 0.0249, J = 0.887$), Two-Stage Relay Optimum ($J_{2\mathrm{stage}}^* = 0.850$), and Unconstrained Adjoint Optimal Control ($J_{\mathrm{PMP}}^* = 0.726$). Across $k_{\mathrm{rep,adv}} \in \{1.00, 1.15, 1.32\}$, the threshold policy adapts t_{gate} from 30.42 d ($h(T_{\max}) = 0.0160$) to 31.68 d (0.0198) and 33.30 d (0.0249).

Reconciling Hoitzing et al. [13] and Mapping the 2D (W_0, k_{dmg}) Controllability Boundary. Hoitzing et al. [13] studied mtDNA populations in which mutants retain partial OXPHOS capacity ($\eta_D = 0.65$), lack replicative advantage ($k_{\mathrm{rep,adv}} = 1.00$), and experience negligible ROS conversion ($k_{\mathrm{dmg}} = 0, B_0 = 0.02$). Evaluating this depleted point ($W_0 = 0.17, D_0 = 0.38$, initial $\mathrm{ATP}(0) = 0.474 > \theta_{\mathrm{surv}} = 0.450$), immediate Purge-First removes ATP-producing D faster than basal biogenesis replaces W, dipping transiently to $\mathrm{ATP}_{\min} = 0.440 < 0.450$, whereas an 18-day Build-First pulse preserves $\mathrm{ATP}_{\min} = 0.474 \ge 0.450$.

To establish the global topological boundary between these regimes—and expose where *no* controller can prevent bioenergetic collapse—we integrate both policies across a 31×31 grid (961 grid points, 1,922 ODE runs) spanning $W_0 \in [0.12, 0.42]$ and $k_{\mathrm{dmg}} \in [0.000, 0.045]$ $\mathrm{d}^{-1}\mathrm{EP}^{-1}$ ($D_0 = 0.38, B_0 = 0.15, \eta_D = 0.65, k_{\mathrm{rep,adv}} = 1.15$). Figure 2B reveals four distinct dynamical zones meeting at a codimension-2 organizing point near $(W_0, k_{\mathrm{dmg}}) \approx (0.185, 0.023)$:

1. **Zone 0 — Irreversible Bioenergetic Collapse (182/961 = 18.9%):** At severely depleted W_0, initial energetic deficit violates $\mathrm{ATP}(t) \ge \theta_{\mathrm{surv}} = 0.450$ or prevents recovery ($W(T_{\max}) < 0.85$) under *both* Build-First and Purge-Then-Build, defining the fundamental controllability limit.

2. **Zone 1 — Hoitzing Build-First Exclusive Rescue (28/961 = 2.9%):** A narrow low-W_0, weak-ROS wedge ($W_0 \in [0.15, 0.19], k_{\mathrm{dmg}} \le 0.023$) where Build-First preserves $\mathrm{ATP}(t) \ge 0.450$ while Purge-First transiently starves ATP.

3. **Zone 2 — Purge-Then-Build Exclusive Rescue (59/961 = 6.1%):** At stronger ROS coupling ($k_{\mathrm{dmg}} \ge 0.023, W_0 \in [0.18, 0.28]$), Build-First drives $\mathrm{EP}(t) > \theta_{\mathrm{GSH}}$ and triggers runaway wild-type loss, whereas Purge-Then-Build rescues the cell.

4. **Zone 3 — Both Safe, Purge-Then-Build Cost-Optimal (692/961 = 72.0%):** Both policies satisfy $\mathrm{ATP}(t) \ge 0.450$, with Purge-Then-Build achieving strictly lower Bolza cost.

3 Accumulated-Selection Reduction and Multi-Term Perturbation Bound

A natural question is whether Stage-1 heteroplasmy clearance across biological systems with vastly different turnover rates (hours in activated T cells vs. weeks in flight muscle vs. months in post-mitotic neurons) can be analyzed in a common dimensionless coordinate. As noted in control-theoretic critiques of naive Buckingham Π reductions [16], dimensional analysis alone cannot reduce an 8-state ODE to a 1D curve unless the time-variation of the contraction rate $\lambda_h(t)$ in Eq. (20) is explicitly accounted for. We therefore establish both (i) the exact accumulated-selection identity and (ii) a complete multi-term perturbation bound for constant-rate rescaling.

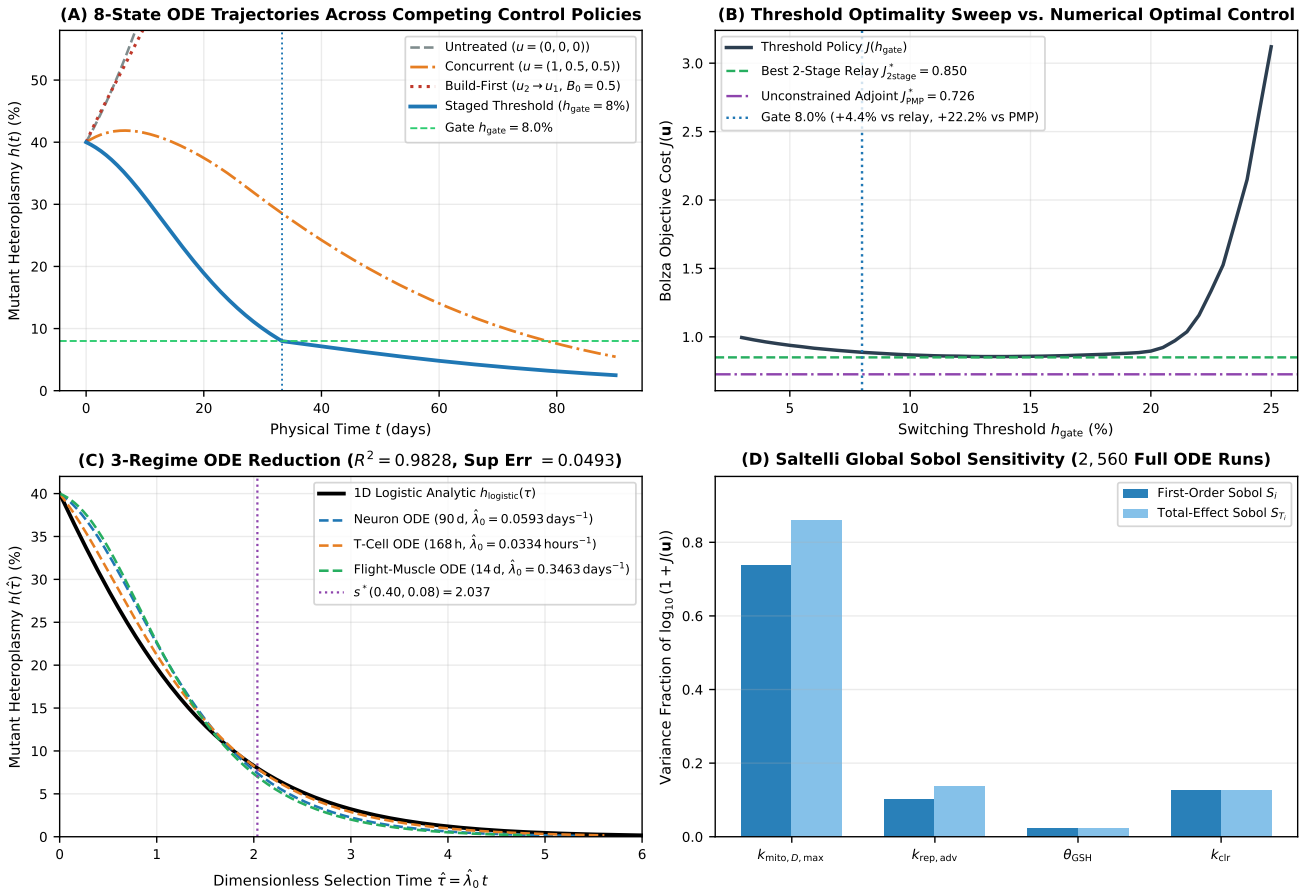


Figure 1: **First-Principles 8-State ODE Control Comparison, Threshold Optimality Sweep, Independent 3-Regime Reduction, and Saltelli Sobol Sensitivity Analysis** (generated by `simulator/generate_paper1_verified_benchmarks.py`, seed `20261109`). **(A)** Mutant heteroplasmy trajectories $h(t)$ under Untreated, Concurrent, Build-First ($B_0 = 0.50$), and Staged Threshold ($h_{\text{gate}} = 8.0\%, B_{\text{gate}} = 0.120$, switching at $t_{\text{gate}} = 33.30$ d) policies. **(B)** Direct numerical sweep of $J(h_{\text{gate}})$ across $h_{\text{gate}} \in [3\%, 25\%]$ compared against both the optimal 2-stage constant relay $J^*_{\text{2stage}} = 0.850$ (+4.38%) and unconstrained 300-variable discrete-adjoint optimal control $J^*_{\text{PMP}} = 0.7262$ (+22.21%). **(C)** Independent 8-state ODE trajectories for Neuronal (90 d), T-Cell (168 h), and Flight-Muscle (14 d) regimes with distinct Π-groups plotted against closed-form constant-rate dimensionless time $\hat{\tau} = \hat{\lambda}_0 t$ ($R^2 = 0.9828$, supremum error 0.0493) and exact accumulated selection $s^* = 2.0369$. **(D)** Global variance-based Sobol sensitivity indices (S_i, S_{T_i}) across 2,560 full ODE integrations.

Proposition 2 (Accumulated-Selection Reduction) *Suppose* $\text{EP}(t) \le \theta_{\text{GSH}}$ *on* $[0, t]$ *in Stage 1* ($u_0 = 1$, $u_1 = 1$, $u_2 = 0$) *with* $h_0 \in (0, 1)$.

1. ***Exact Accumulated-Selection Coordinate*** $s(t)$***:*** *Define the dimensionless accumulated selection integral*

$$s(t) \triangleq \int_0^t \lambda_h(q)\, dq. \tag{30}$$

Then the 8-state ODE heteroplasmy trajectory $h(t)$ *satisfies the exact 1D logistic identity*

$$h(t) = h_{\text{logistic}}\big(s(t); h_0\big) \triangleq \frac{h_0\, e^{-s(t)}}{1 - h_0 + h_0\, e^{-s(t)}}, \tag{31}$$

and reaches any target threshold $h_{\text{gate}} \in (0, h_0)$ *at the exact accumulated-selection value*

$$s^*(h_0, h_{\text{gate}}) = \ln\!\left(\frac{h_0(1 - h_{\text{gate}})}{h_{\text{gate}}(1 - h_0)}\right), \tag{32}$$

which depends explicitly on (h_0, h_{gate}) *(giving* $s^*(0.40, 0.08) = \ln(4.6/0.6) = 2.03688$, $s^*(0.25, 0.08) = 1.34373$, *and* $s^*(0.60, 0.08) = 2.84781$*).*

2. ***Zero-Tuning Closed-Form Perturbation Bound for Constant-Rate Rescaling*** $\hat{\tau} = \hat{\lambda}_0 t$***:*** *Because* $u_2 \equiv 0$ *during Stage 1, Eq.* (3) *gives* $k_{\text{bio}}(0, N) \equiv k_{\text{bio},0}$ *identically (zero cofactor rate error), while* $\dot{B} = b_{\text{basal}} - k_{\text{clr}} B$ *integrates in closed form to* $B(q) = B_{\text{eq}} + (B_0 - B_{\text{eq}})e^{-k_{\text{clr}} q}$. *Let* $T_{1,\text{apriori}} \triangleq s^*(h_0, h_{\text{gate}}) / \big[k_{\text{mito},D,\max} V_{\text{lys}}(B_{\text{eq}}) -$

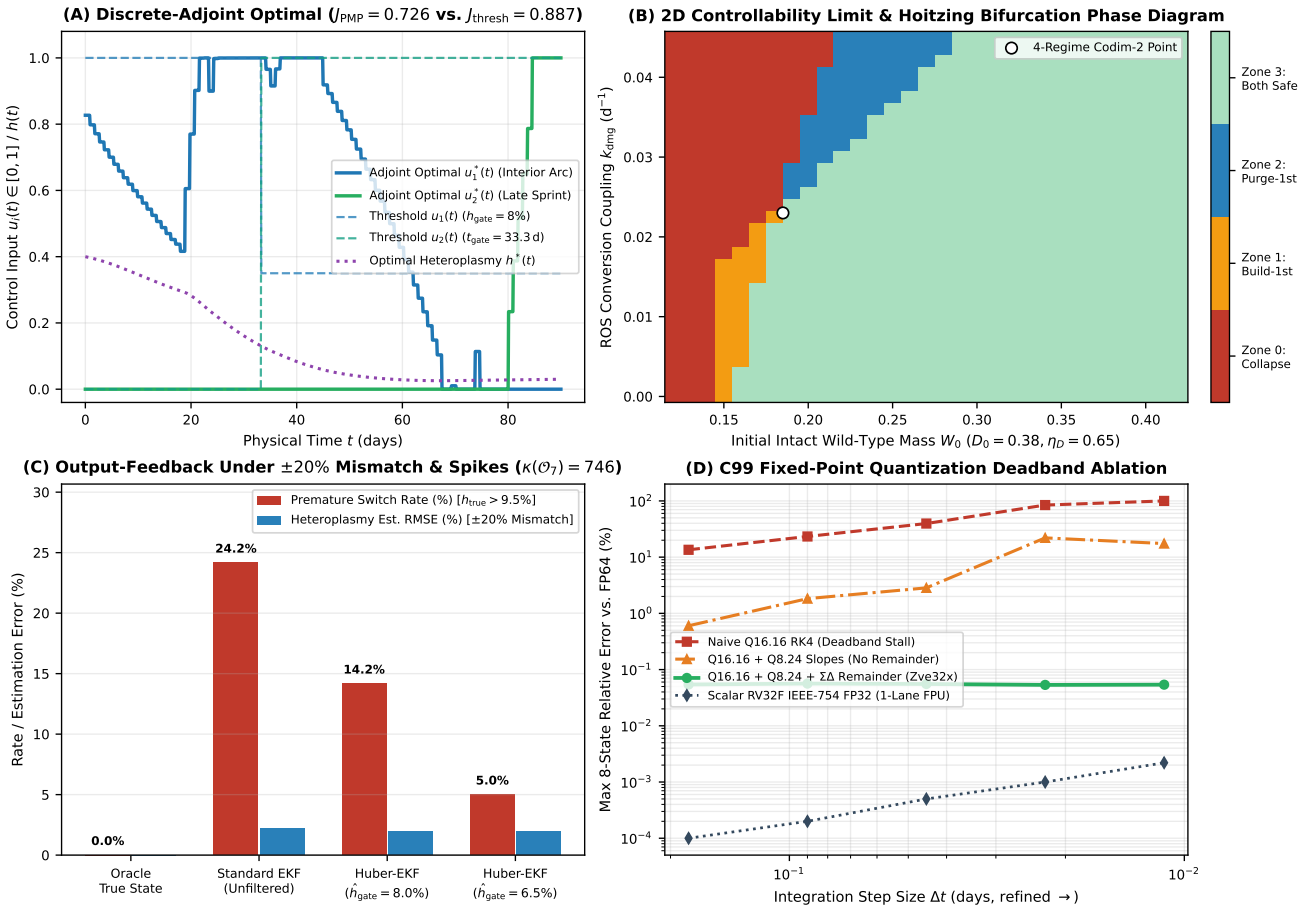


Figure 2: **Unconstrained Discrete-Adjoint Optimal Control, 2D Controllability Phase Diagram, Mismatched Closed-Loop Huber-EKF Audit, and C99 Fixed-Point Deadband Ablation.** **(A)** 300-variable unconstrained `L-BFGS-B` discrete-adjoint optimal controls $(u_1^*(t), u_2^*(t))$ ($J_{\mathrm{PMP}}^* = 0.7262$, 5-start spread 0.12%) vs. closed-loop staged threshold control ($J_{\mathrm{thresh}} = 0.8875$), showing the interior redox-boundary modulation of $u_1^*(t)$ and late sprint of $u_2^*(t)$. **(B)** 31×31 phase diagram over (W_0, k_{dmg}) (961 points, $D_0 = 0.38, \eta_D = 0.65$), mapping Zone 0 Irreversible Collapse (18.9%), Zone 1 Build-First Exclusive Rescue (2.9%), Zone 2 Purge-First Exclusive Rescue (6.1%), and Zone 3 Both Safe (72.0%). **(C)** 120-trial closed-loop Output-Feedback Monte Carlo benchmark (1,000 steps, $\Delta t = 0.09$ d) under ±20% 7-rate mismatch, ±8% initial-state uncertainty, AR(1) noise ($\rho = 0.60$), and 5% 6.5σ spikes ($\kappa(\mathcal{O}_7) = 745.6$). **(D)** Multi-step-size C99 quantization deadband ablation ($\Delta t \in [0.18, 0.01125]$ d) comparing naive `Q16.16 RK4`, `Q8.24` slopes without remainder, `Q8.24` slopes with 8-lane ΣΔ remainder diffusion, and scalar `RV32F FP32`.

$k_{\mathrm{mito},W} - (k_{\mathrm{rep,adv}} - 1)k_{\mathrm{bio},0}(1 - M_0/K)\big]$ *be the cleared-bottleneck horizon estimate, and define the a priori Stage-1 mean lysosomal competence* $\bar{V}_{\mathrm{lys}} = V_{\mathrm{lys}}(\bar{B})$ *with zero empirical tuning constants:*

$$\bar{V}_{\mathrm{lys}} \triangleq \frac{1}{T_{1,\mathrm{apriori}}} \int_0^{T_{1,\mathrm{apriori}}} \frac{K_B\, dq}{K_B + B(q)} = \frac{K_B}{K_B + B_{\mathrm{eq}}} \left[1 - \frac{\ln\!\left(\frac{K_B + B_0}{K_B + B(T_{1,\mathrm{apriori}})}\right)}{k_{\mathrm{clr}}\, T_{1,\mathrm{apriori}}} \right], \quad (33)$$

$$\hat{\lambda}_0 \triangleq k_{\mathrm{mito},D,\max}\, \bar{V}_{\mathrm{lys}} - k_{\mathrm{mito},W} - (k_{\mathrm{rep,adv}} - 1)k_{\mathrm{bio},0} \left(1 - \frac{M_0}{K}\right). \quad (34)$$

Let $\hat{\tau} = \hat{\lambda}_0 t$, $\ell(t) = \ln \frac{h(t)}{1-h(t)}$, *and* $\hat{\ell}(\hat{\tau}) = \ln \frac{h_0}{1-h_0} - \hat{\tau}$. *Then for all* $t \in [0, t_{\mathrm{gate}}]$, $|\ell(t) - \hat{\ell}(\hat{\lambda}_0 t)| = |s(t) - \hat{\lambda}_0 t|$ *satisfies the two-term perturbation inequality:*

$$|s(t) - \hat{\lambda}_0 t| \le \underbrace{k_{\mathrm{mito},D,\max} \int_0^t \left| V_{\mathrm{lys}}(B(q)) - \bar{V}_{\mathrm{lys}} \right| dq}_{\mathcal{E}_B(t)\ \text{(bottleneck)}} + \underbrace{(k_{\mathrm{rep,adv}} - 1) \frac{k_{\mathrm{bio},0}}{K} \int_0^t \left| M(q) - M_0 \right| dq}_{\mathcal{E}_M(t)\ \text{(mass, >0 at } B_0\text{=0)}}, \quad (35)$$

and since $\left| \frac{d}{d\ell} \frac{e^\ell}{1+e^\ell} \right| = h(1-h) \le \frac{1}{4}$, *the heteroplasmy supremum error satisfies*

$$\left| h(t) - h_{\mathrm{logistic}}(\hat{\lambda}_0 t; h_0) \right| \le \frac{1}{4} \Big(\mathcal{E}_B(t) + \mathcal{E}_M(t) \Big). \quad (36)$$

Proof. Part 1 follows by integrating $\frac{d}{dt}\ln\left(\frac{h}{1-h}\right) = -\lambda_h(t)$ over $[0,t]$ and inverting the logit map. For Part 2, substituting $u_1 = 1, u_2 = 0$ into Eq. (20) eliminates $N(q)$ identically; integrating $V_{\rm lys}(B(q))$ via $w = e^{k_{\rm clr}q}$ yields Eq. (33). Subtracting Eq. (34) from Eq. (20), applying the triangle inequality (noting also $|V_{\rm lys}(B) - V_{\rm lys}(\bar{B})| \le |B - \bar{B}|/K_B$), and bounding the logistic derivative by $\max_{h\in[0,1]} h(1-h) = 1/4$ proves Eqs. (35)–(36). Notably, $\mathcal{E}_M(t)$ remains strictly positive even when $B_0 = B_{\rm eq} = 0$. ■

Table 2 and Figure 1C verify Proposition 2 across three genuinely non-homothetic regimes (not scalar time-multiples of one parameter vector). At $h_0 = 0.40$ (where $\mathrm{EP}(t) \le \theta_{\rm GSH}$ holds throughout Stage 1), the exact $s(t)$ coordinate achieves 0.0000 error, and the zero-tuning closed-form rescaling $\hat{\tau} = \hat{\lambda}_0 t$ satisfies Eq. (36) in every regime ($0.0422 \le 0.1216$, $0.0216 \le 0.0399$, $0.0493 \le 0.1213$; pooled $R^2 = 0.9828$). Conversely, the severe perturbation $h_0 = 0.60^*$ ($W_0 = 0.40, D_0 = 0.60$) illustrates the exact failure boundary of 1D sub-damage reduction: initial $\mathrm{EP}(0)$ breaches $\theta_{\rm GSH} = 1.150$ ($\mathrm{EP}_{\rm peak} = 9.041, 1.312, 9.734$), activating positive ROS conversion $+k_{\rm dmg}(1-h)(\mathrm{EP} - \theta_{\rm GSH})$ in Eq. (19) and delaying $\hat{\tau}_{\rm gate}$ to 3.024–$5.336 > s^*(0.60, 0.08) = 2.848$.

4 Embedded Fixed-Point Implementation on Coral NPU

4.1 Target Architecture and Single-Bank L1 DTCM Memory Budget

We target the open-source Google Coral NPU architecture (`RV32IMF_Zve32x`) [17, 18], which combines a 32-bit RISC-V scalar core with a 128-bit single-instruction multiple-data (SIMD) vector execution unit (`Zve32x` extension supporting four 32-bit lanes or eight 16-bit lanes per register), 8 KB of instruction tightly coupled memory (ITCM), and 32 KB of data tightly coupled memory (DTCM) organized as four 8,192-byte SRAM banks.

To minimize static leakage power during always-on chronic bioelectronic monitoring, our embedded design constrains the entire runtime data footprint (INT8 projection LUTs, mixed `Q16.16`/`Q8.24` 7-state Huber-EKF covariance matrices, `RK4` slope registers, 64-frame sensor ring buffers, and scalar/ISR stack) to a static C99 struct `coralnpu_dtcm_bank0_workspace_t` verified at compile and run time to occupy 3,696 **B (45.12%) of a single 8,192-byte L1 DTCM bank (`Bank 0`, leaving 4,496 B free headroom)**, allowing the remaining three 8 KB DTCM banks (24 KB) to remain power-gated in deep retention mode (Figure 3D and Table 6).

4.2 Scalar `FP32` vs. `Zve32x` Integer SIMD, Fixed-Point Huber-EKF, and Deadband-Free `RK4`

An explicit architectural nuance of `RV32IMF_Zve32x` merits clarification: the scalar RISC-V core includes a single-lane IEEE-754 single-precision floating-point unit (`F` extension, `FP32`), whereas its 128-bit vector unit implements **integer-only** `Zve32x` (supporting four parallel 32-bit integer lanes or eight 16-bit lanes, without vector floating-point `Zve32f`). Consequently, scalar single-compartment integration can execute directly in `FP32`, whereas vectorizing multi-lane state updates across the 128-bit SIMD datapath requires signed 32-bit `Q16.16` fixed-point arithmetic ($1.0 = 65{,}536$, resolution $2^{-16} \approx 1.526 \times 10^{-5}$).

In `build/coralnpu/mitorestorer_coralnpu_kernel.h`, exact firmware parity across multi-week biological timescales combines three mechanisms:

1. **Bit-Exact 64-Bit Widening Vector Stage Updates:** Multiplying `Q16.16` operands or scaling `Q8.24` slopes by $\Delta t_{\rm Q16}$ produces 64-bit intermediates (`Q32.32` or `Q24.40`). Both `coralnpu_q24_axpy8_q16` (inside our 4-stage `RK4` stepper) and `coralnpu_rvv_rk4_bioenergetic_step` execute 32-to-64-bit widening vector multiplication (`__riscv_vwmul_vx_i64m4`) followed by narrowing arithmetic right shift (`__riscv_vnsra_wx_i32m2(..., 24)` or `16`). Across 256 randomized 8-lane trials (2,048 lane operations), widening vector execution achieves **0-ULP discrepancy** against scalar 64-bit arithmetic.

2. **`Q8.24` Slope Scaling and 8-Lane ΣΔ Sub-LSB Remainder Diffusion:** Because slow biological rates ($k_{{\rm mito},W} = 0.015\ \mathrm{d}^{-1}$) multiplied by refined step sizes $\Delta t \le 0.045$ d produce per-step increments smaller than 0.5 LSB_{16} (7.63×10^{-6}), naive `Q16.16` Runge-Kutta integration suffers complete quantization deadband stalling as $\Delta t \to 0$ (Figure 2D). Even evaluating stage derivatives $\mathbf{k}_1 \ldots \mathbf{k}_4$ in left-shifted `Q8.24` format ($2^{-24} \approx 5.96 \times 10^{-8}$) still hits the ± 0.5 LSB_{16} state-update rounding deadband when Δt is small. To break this barrier, `coralnpu_rk4_core8_full_step_q16` accumulates the exact 64-bit weighted numerator $\Delta t \cdot (\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4) + \mathbf{r}_{k-1}$ against divisor $6 \times 2^{24} = 100{,}663{,}296$ and propagates the signed sub-LSB remainder vector $\mathbf{r}_k \in (-100{,}663{,}296,\ 100{,}663{,}296)^8$ across steps as a first-order ΣΔ error-diffusion accumulator.

3. **Bare-Metal Mixed `Q16.16`/`Q8.24` 7-State Sequential Scalar Huber-EKF:** To execute closed-loop estimation without dynamic memory (`malloc`) or `<math.h>` floating-point library calls, `coralnpu_huber_ekf7_meas_update_q16` stores state estimates $\hat{\mathbf{x}}_{0:6}$ in `Q16.16` and covariance

Table 2: Non-Trivial Three-Regime 8-State ODE Verification of Proposition 2 Across Distinct Dimensionless Π-Groups (`simulator/generate_paper1_verified_benchmarks.py`). Each regime has distinct ratios ($\Pi_1 = k_{\mathrm{mito},W}/k_{\mathrm{mito},D,\max}$, $k_{\mathrm{rep,adv}}$, K, K_B, B_0, η_D), uses the zero-tuning closed-form rate $\hat{\lambda}_0$ in Eqs. (33)–(34), and verifies that the Stage-1 supremum error stays strictly below the analytical bound $\frac{1}{4}(\mathcal{E}_B + \mathcal{E}_M)$ of Eq. (36).

ODE Regime (Horizon $T_{\max}$)	**Distinct Π-Groups** $(\Pi_1, k_{\mathrm{rep}}, K, K_B, B_0, \eta_D)$	$\hat{\lambda}_0$ (unit^{-1})	t_{gate} ($h_0 = 0.40$)	$\hat{\tau}_{\mathrm{gate}} = \hat{\lambda}_0 t_{\mathrm{gate}}$ ($h_0 = 0.40/0.25/0.60^*$)	$h(\hat{\tau})/h_0$ at $\hat{\tau} = 2.0/4.0/6.0$	**Exact $s(t)$ Sup Err**	**Stage-1 Sup Err vs. Prop. 2 Bound**	**Full Const-Rate Sup Err / RMSE**	**Const-Rate** R^2
1. Neuron (90.0 d)	$(0.130, 1.32, 1.25, 0.25, 0.50, 0.15)$	0.05928 d^{-1}	33.30 d	1.974 / 1.414 / 5.336	0.1934 / 0.0153 / 0.0026	0.0000	$0.0422 \le 0.1216$	0.0422 / 0.0168	0.9812
2. CAR-T (168.0 h)	$(0.125, 1.12, 1.40, 0.35, 0.28, 0.28)$	0.03340 h^{-1}	60.82 h	2.031 / 1.403 / 3.024	0.2062 / 0.0241 / 0.0040	0.0000	$0.0216 \le 0.0399$	0.0216 / 0.0084	0.9946
3. *Apis* (14.0 d)	$(0.125, 1.22, 1.18, 0.20, 0.60, 0.08)$	0.34632 d^{-1}	5.57 d	1.930 / 1.396 / 4.848	0.1832 / 0.0129 / 0.0040	0.0000	$0.0493 \le 0.1213$	0.0493 / 0.0203	0.9745
1D Analytic Logistic	$\mathrm{EP}(t) \le \theta_{\mathrm{GSH}}$ ($s^* = \hat{\tau}_{\mathrm{gate}}$)	1.00000	2.0369	2.037 / 1.344 / 2.848	0.2069 / 0.0302 / 0.0041	—	**All 3 Hold**	**Pooled:** 0.0493 / 0.0160	**0.9828**

$\mathbf{P} \in \mathbb{Z}_{32}^{7\times 7}$ in `Q8.24` (preserving variance resolution down to $2^{-24} \approx 6 \times 10^{-8}$), processes the four channels of $\mathbf{y}_k$ sequentially via scalar innovation variances $S_i = \mathbf{h}_i \mathbf{P} \mathbf{h}_i^\top + R_i$ (avoiding 4×4 matrix inversion), extracts innovation standard deviations via exact 64-bit integer Newton square root (`coralnpu_isqrt64`), and applies symmetric Huber weight scaling $w_{\mathrm{Hub}} = \min(1, \gamma_{\mathrm{Huber}}\sqrt{S_i}/|\nu_i|)$ directly to both state and covariance updates.

5 Output-Feedback Observability, Disturbance Rejection, and Verification

5.1 64-Frame Dual-Modality Persistence Filter Under Markov Correlation

To prevent transient sensor artifacts from prematurely triggering Stage-2 biogenesis or emergency rollback, the controller evaluates a 64-frame sliding bitmask register $\mathcal{B}_k \in \{0,1\}^{64}$ on primary stress channel $y_{k,1}$ with quorum threshold $M_{\mathrm{req}} = 60$, gated by a concurrent secondary modality check $z_k = y_{k,2} \ge \gamma_{\mathrm{sec}}$ (`coralnpu_rolling_fuzzy_sensor_gate`).

Proposition 3 (Markov Persistence Bound) *Suppose stationary sensor noise evolves via the reversible Gaussian* AR(1) *Markov chain* $e_k = \rho\, e_{k-1} + \sqrt{1-\rho^2}\,\epsilon_k$ *on* $\mathbb{R}$ *(*$\epsilon_k \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$*,* $\rho \in [0,1)$*, stationary measure* $\pi = \mathcal{N}(0,1)$*), whose Markov operator possesses* $L^2(\pi)$ *absolute spectral gap* $\gamma_{\mathrm{spec}} = 1 - \rho > 0$*. Let* $I_k = f(e_k) \triangleq \mathbb{I}(y_{k,1}(e_k) \ge \gamma_{\mathrm{crit}}) \in \{0,1\}$ *be the bounded measurable primary exceedance indicator with marginal expectation* $p_y = \mathbb{E}_\pi[I_k] < \mu_{\mathrm{req}} \triangleq M_{\mathrm{req}}/N_{\mathrm{win}} = 60/64 = 0.9375$*, let* $p_{z|Q} \triangleq \mathbb{P}\big(z_k \ge \gamma_{\mathrm{sec}} \,\big|\, \sum_{j=k-63}^{k} I_j \ge 60\big)$ *denote the quorum-conditioned secondary exceedance probability, and let* $c_0 \in \{0, \dots, 64\}$ *denote the exceedance count in the 64-frame ring buffer immediately prior to a contiguous deterministic primary disturbance of length* L_{dist} *frames. Then:*

1. ***Deterministic Sub-Quorum Rejection Condition:*** *Reaching primary quorum* $\sum_{j=k-63}^{k} I_j \ge 60$ *during the disturbance is deterministically impossible whenever*
$$L_{\mathrm{dist}} \le 59 - c_0. \tag{37}$$
Under a clean prior buffer ($c_0 = 0$)*, zero false triggers occur for* $L_{\mathrm{dist}} \le 59$ *frames; with* $c_0 > 0$ *prior noise exceedances, quorum requires* $L_{\mathrm{dist}} \ge 60 - c_0$ *(Figure 3B).*

2. ***Spectral-Gap Concentration and Horizon Union Bound:*** *Unlike classical Hoeffding bounds requiring independent samples [19], applying Paulin's concentration inequality for bounded functions* $f : \mathbb{R} \to [0,1]$ *of reversible Markov chains [20] to* e_k *bounds the single-step and* K_{frames}*-horizon false-trigger probabilities by:*
$$P_{\mathrm{FP,step}} \le p_{z|Q}\, \exp\!\left(-2\, N_{\mathrm{win}}\, \frac{1-\rho}{1+\rho}\, (\mu_{\mathrm{req}} - p_y)^2\right), \tag{38}$$
$$P_{\mathrm{FP,horizon}} \le \min\big(1,\, K_{\mathrm{frames}}\, P_{\mathrm{FP,step}}\big). \tag{39}$$

Proof. Any 64-frame window overlapping a contiguous L_{dist}-frame burst contains at most $c_0 + L_{\mathrm{dist}}$ ones, which is < 60 whenever $L_{\mathrm{dist}} \le 59 - c_0$, proving Eq. (37). Note that binary thresholded indicators I_k of an AR(1) process are not themselves a 2-state Markov chain, whereas the latent Gaussian process e_k *is* a stationary reversible Markov chain on $(\mathbb{R}, \pi)$ with Mehler–Hermite spectrum $\{\rho^m\}_{m\ge 1}$ and $L^2(\pi)$ spectral gap $\gamma_{\mathrm{spec}} = 1 - \rho$. Applying Theorem 3.3 of Paulin [20] directly to the bounded measurable function $f(e_k) \in [0,1]$ yields $\mathbb{P}_\pi(\sum_{j=1}^{64} f(e_j) \ge 60) \le \exp(-2N_{\mathrm{win}}\frac{1-\rho}{1+\rho}(\mu_{\mathrm{req}} - p_y)^2)$. Multiplying by the exact conditioning factor $p_{z|Q}$ and applying Boole's union bound over K_{frames} steps yields Eqs. (38)–(39). ∎

Explicit Frame Periods Across Deployments. In physical time, a 64-frame window spans $T_{\mathrm{win}} = 64\,\Delta t_{\mathrm{frame}}$, where $\Delta t_{\mathrm{frame}}$ matches the plant timescale: $\Delta t_{\mathrm{frame}} = 1.0$ s for *Apis* acoustic telemetry ($T_{\mathrm{win}} = 64$ s, using a 16-bin Goertzel/FFT over 128 ms at 1 kHz for 7.81 Hz bin resolution between 220 Hz and 240 Hz), $\Delta t_{\mathrm{frame}} = 1.0$ min for CAR-T impedance ($T_{\mathrm{win}} = 64$ min), and $\Delta t_{\mathrm{frame}} = 15.0$ min for wearable/implantable neural monitoring ($T_{\mathrm{win}} = 16.0$ h). Table 3B reports the 1,000-stream Monte Carlo evaluation ($K_{\mathrm{frames}} = 1{,}000$, seed `20261109`) with exact Clopper-Pearson 95% upper confidence bounds.

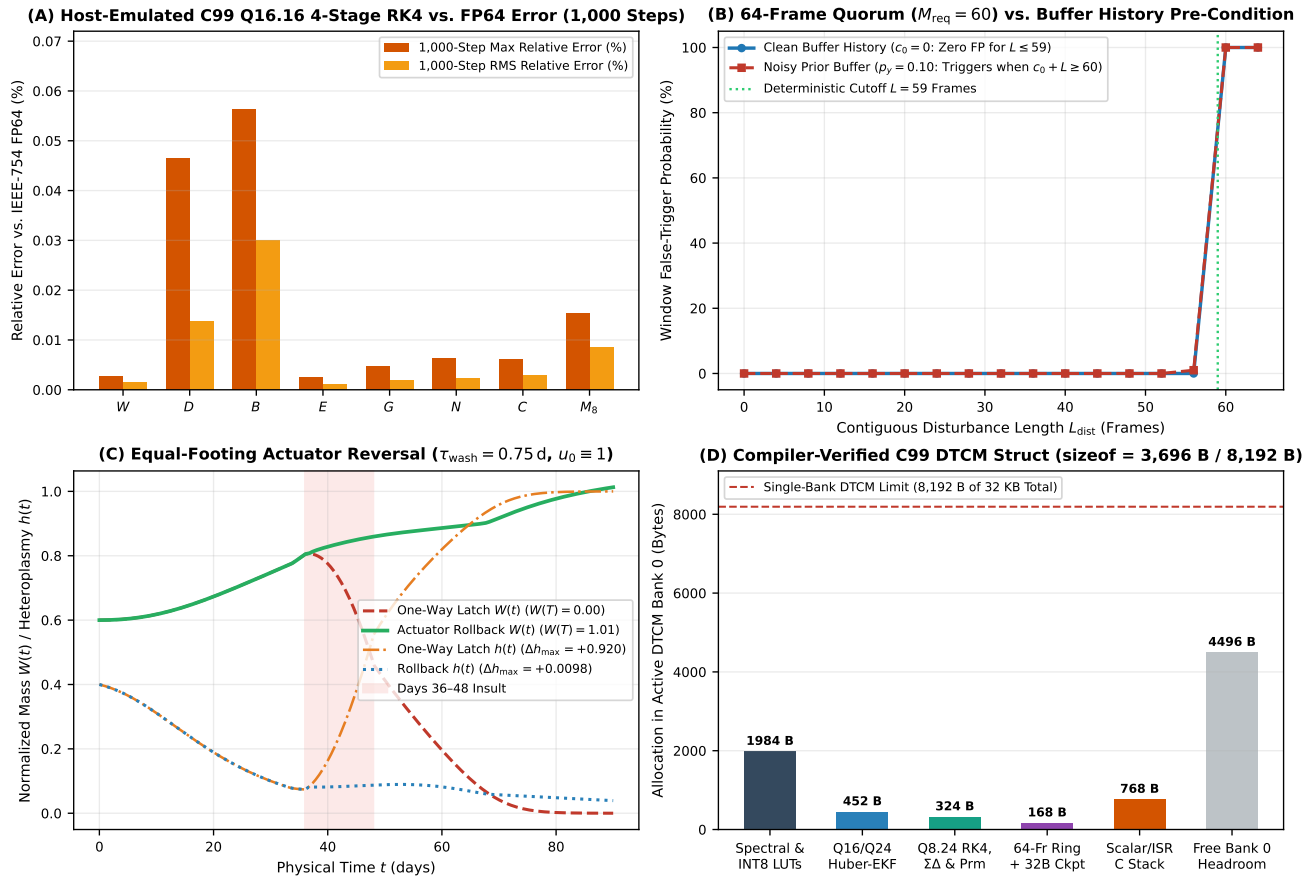


Figure 3: **Host-Emulated C99 `Q16.16` 4-Stage `RK4` Parity, 64-Frame Quorum Pre-Condition Analysis, Equal-Footing Plant-Preserving Rollback, and Verified L1 DTCM Footprint.** **(A)** Full-horizon (1,000-step) maximum and RMS relative error (%) between the bare-metal C99 `Q16.16` 4-stage `RK4` kernel (`coralnpu_hil_testbench.c`) and IEEE-754 `FP64` across all 8 states (W, D, B, E, G, N, C, M_8; global max error 0.0563%). **(B)** 64-frame persistence false-trigger rate ($M_{\text{req}} = 60$) vs. contiguous disturbance length L_{dist} under clean prior buffer history ($c_0 = 0$: zero false triggers for $L_{\text{dist}} \leq 59$) vs. noisy prior buffer ($p_y = 0.10$). **(C)** Equal-footing ($u_0 \equiv 1$) plant-preserving actuator reversal under an acute Days 36–48 exogenous surge ($\tau_{\text{wash}} = 0.75$ d): the biological plant state $\mathbf{x}(t)$ evolves continuously without state reset, bounding $\Delta h_{\max} = +0.0098$ and rescuing $W(T_{\max}) = 1.0128$ vs. bioenergetic collapse ($W \to 0.0001$) under a one-way Stage-2 latch. **(D)** Compiler-verified static L1 DTCM workspace (`sizeof(coralnpu_dtcm_bank0_workspace_t)` = 3,696 B within the 8,192 B Bank 0 budget).

Table 3: First-Principles Policy Comparison (Aligned 1,000-Interval Bolza Integration, Including Multi-Start 300-Variable Discrete-Adjoint Optimum) and 1,000-Stream Correlated AR(1) Persistence Benchmarks (`simulator/generate_paper1_verified_benchmarks.py`).

A. Neuronal 8-State Control Policy ($T_{\max} = 90$ **d**)	**Bolza Cost** $J(\mathbf{u})$	**Switch** t_{gate}	**Terminal** $h(T_{\max})$	**Terminal** $W(T_{\max})$	**Peak** $\text{EP}_{\max}$	**Min** $\text{ATP}_{\min}$
Untreated Baseline ($\mathbf{u} = (0, 0, 0)$)	2,768,855.31	—	1.0000	0.0000	20.605	0.186 (Violated)
Concurrent Biogenesis + Mitophagy ($\mathbf{u} = (1, 0.5, 0.5)$)	99.829	—	0.0546	0.9474	1.423	0.806 (Safe)
Build-First Policy in High-Impedance Regime ($B_0 = 0.50$)	760,388.40	—	1.0000	0.0000	9.744	0.028 (Violated)
Proposed Staged Threshold Policy ($h_{\text{gate}} = 0.080, B_{\text{gate}} = 0.120$)	**0.8875**	**33.30 d**	**0.0249**	**1.0487**	**1.046**	**0.806 (Safe)**
Two-Stage Constant Relay Optimum ($t^*_{\text{sw}} = 48.18$ d, $u^*_{1,S2} = 0$)	0.8500 (J^*_{2stage})	48.18 d	0.0375	1.0421	1.046	0.806 (Safe)
Unconstrained 300-Var Discrete-Adjoint Optimum (`L-BFGS-B`)	0.7262 (J^*_{PMP})	Interior Arc	0.0300	0.9938	1.150	0.806 (Safe)
B. 1,000-Stream Correlated AR(1) **Noise Benchmark**	**Lag-1** ρ	**Burst** L_{dist}	**Single-Sample FP**	**64-Quorum Primary FP**	**64-Quorum Dual FP**	**95% Upper Bound**
1. *Apis* Wind-Gust Disturbance ($\Delta t_{\text{frame}} = 1$ s)	0.50	12 frames	1000/1000 (100.0%)	0/1000 (0.00%)	0/1000 (0.00%)	$\leq 0.299\%$
2. CAR-T Lactate Biofouling Spike ($\Delta t_{\text{frame}} = 1$ min)	0.70	25 frames	1000/1000 (100.0%)	0/1000 (0.00%)	0/1000 (0.00%)	$\leq 0.299\%$
3. Wearable Postprandial Excursion ($\Delta t_{\text{frame}} = 15$ min)	0.60	45 frames	1000/1000 (100.0%)	11/1000 (1.10%)	7/1000 (0.70%)	$\leq 1.219\%$

5.2 Stage-1 Observability Gramian and Closed-Loop Huber-EKF Under ±20% Mismatch

Because state M_8 is a purely downstream epigenetic readout ($\partial \dot{\mathbf{x}}_{0:6}/\partial M_8 = \mathbf{0}$, $\partial \mathbf{h}_{\text{obs}}/\partial M_8 = \mathbf{0}$, giving exact

singular value $\sigma_8 = 0$), output-feedback estimation operates on the 7-state core $\mathbf{x}_{0:6} = [W, D, B, E, G, N, C]^\top$ from the 4-channel proxy $\mathbf{y}_k$ in Eq. (18). Because the switching decision must be made *prior* to Stage-2 excitation, we evaluate the empirical local Observability Gramian $\mathcal{O}_7$ strictly over the pre-switch Stage-1 window $t \in [0, 33.30\ \mathrm{d}]$ (370 steps at the exact controller sampling step $\Delta t = 0.09$ d), yielding singular values $\sigma(\mathcal{O}_7) = [963.44, 253.33, 205.73, 171.19, 137.38, 15.61, 1.29]$ and condition number $\kappa(\mathcal{O}_7) = \sigma_1/\sigma_7 = 745.6$. Consistent with §2.1, the weakest observable mode ($\sigma_7 = 1.29$, eigenvector $0.980C - 0.162W - 0.104D$) aligns with multiplicative iso-ATP compensation in $(W + \eta_D D)(1 + \gamma_C C)$, while the second-weakest mode ($\sigma_6 = 15.61$, eigenvector $0.978B - 0.193D$) reflects (B, h) collinearity in $1 + \beta_B B + \beta_h h$.

We execute a 1,000-step ($\Delta t = 0.09$ d), 120-trial Monte Carlo benchmark in which every true plant independently draws uniform ±20% perturbations across all 7 core kinetic parameters $(k_{\mathrm{mito},W}, k_{\mathrm{mito},D,\max}, k_{\mathrm{bio},0}, k_{\mathrm{bio},\max}, k_{\mathrm{rep,adv}}, k_{\mathrm{clr}}, K_B)$ plus ±8% initial-state uncertainty unknown to the EKF internal model, corrupted simultaneously by correlated AR(1) sensor noise ($\rho = 0.60$) and 5% heavy-tailed 6.5σ measurement spikes (Figure 2C and Table 4A). Using consistent single-weight Huber M-estimation ($w_{\mathrm{Hub}} = \min(1, \gamma_{\mathrm{Huber}}\sqrt{S_i}/|\nu_i|)$ at $\gamma_{\mathrm{Huber}} = 1.50$) and defining a premature switch as latching Stage 2 while $h_{\mathrm{true}} > 0.095$:

- **Unfiltered Standard EKF ($\hat{h}_{\mathrm{gate}} = 0.080$):** Outlier Kalman innovations and mismatch bias yield 2.18% heteroplasmy RMSE and trigger premature Stage-2 transitions in 29/120 trials (24.17%).

- **Huber-EKF + Persistence ($\hat{h}_{\mathrm{gate}} = 0.080$):** Single-weight Huber innovation clipping ($\gamma_{\mathrm{Huber}} = 1.50$) with 8-step persistence gating lowers RMSE to 1.95% and reduces premature transitions to 17/120 (14.17%), leaving only the steady $\sim 1.9\%$ observability-conditioned mismatch bias.

- **Margin-Tightened Huber-EKF ($\hat{h}_{\mathrm{gate}} = 0.065$):** Tightening the observer switching threshold by 0.015 reduces premature switches by 4.8× relative to Standard EKF to 6/120 (5.00%, $\hat{h}$ RMSE 1.93%), achieving mean terminal $W(T_{\max}) = 1.0479$ and $h(T_{\max}) = 0.0218$ (median $J = 0.933$).

5.3 Equal-Footing Plant-Preserving Actuator Reversal vs. One-Way Latch

A critical distinction in biological control is that **a digital microcontroller can restore its internal reference registers $\mathbf{x}_{\mathrm{ckpt}}$, but it cannot reset the physical biological plant state $\mathbf{x}(t)$**. To evaluate actuator rollback on a strictly equal footing ($u_{0,\mathrm{target}} \equiv 1.0$ identical for both policies), we apply a controller-independent exogenous proteostatic/uncoupling insult during Days 36–48 ($\Delta b_{\mathrm{basal}} = +0.52\ \mathrm{d}^{-1}$ and acute uncoupling load $c_2 = 0.32$):

1. **Detection Latency ($\delta t_{\mathrm{detect}} = 0.54$ d):** Once $\mathrm{EP}(t) > \theta_{\mathrm{abort}} = 1.18$ persists across 6 consecutive controller steps (0.54 d), the controller loads $\mathbf{x}_{\mathrm{ckpt}}$ into its supervisory gating register and commands Stage-2 → Stage-1 actuator reversal ($u_{2,\mathrm{target}} \leftarrow 0, u_{1,\mathrm{target}} \leftarrow 1$, keeping $u_{0,\mathrm{target}} = 1$).

2. **Pharmacokinetic Washout ($\tau_{\mathrm{wash}} = 0.75$ d):** Physical actuation evolves via $\tau_{\mathrm{wash}}\dot{\mathbf{u}}_{\mathrm{eff}} = \mathbf{u}_{\mathrm{target}} - \mathbf{u}_{\mathrm{eff}}$, while the biological plant $\mathbf{x}(t)$ evolves continuously without any discontinuous state reset.

3. **Post-Disturbance Excursion Bound ($\Delta h_{\max}$):** Because ROS conversion is active only while $\mathrm{EP}(t) > \theta_{\mathrm{GSH}}$ during the detection plus washout window $\Delta t_{\mathrm{active}} \approx \delta t_{\mathrm{detect}} + \tau_{\mathrm{wash}} \ln\left(\frac{\mathrm{EP}_{\mathrm{peak}} - \theta_{\mathrm{GSH}}}{\theta_{\mathrm{abort}} - \theta_{\mathrm{GSH}}}\right)$, Eq. (19) bounds the net heteroplasmy excursion by

$$\Delta h_{\max} \le (1 - h_{\mathrm{gate}})\, k_{\mathrm{dmg}} \times \int_{t_{\mathrm{insult}}}^{t_{\mathrm{insult}} + \Delta t_{\mathrm{active}}} \max\big(0, \mathrm{EP}(q) - \theta_{\mathrm{GSH}}\big)\, dq. \tag{40}$$

As shown in Figure 3C, even with full housekeeping clearance $u_0 \equiv 1$, keeping Stage-2 biogenesis latched on ($u_2 = 1$) during the insult drives $\mathrm{EP}_{\mathrm{peak}} = 11.128 > \theta_{\mathrm{GSH}}$ and causes irreversible ROS-driven collapse ($\Delta h_{\max} = +0.9202$, $\mathrm{ATP}_{\min} = 0.138 < \theta_{\mathrm{surv}}$, terminal $W(T_{\max}) = 0.0001$), whereas plant-preserving actuator reversal caps $\mathrm{EP}_{\mathrm{peak}} = 1.521$, bounds $\Delta h_{\max} = +0.0098$ (well below Eq. 40's analytical bound 0.033), maintains $\mathrm{ATP}_{\min} = 0.806 > \theta_{\mathrm{surv}}$, re-enters Stage 2 once $h(t) \le 0.080$, $B(t) \le 0.120$, and $\mathrm{EP}(t) \le 0.950$, and restores $W(T_{\max}) = 1.0128$ and $h(T_{\max}) = 0.0392$.

5.4 Host-Emulated Fixed-Point Parity, C99 Huber-EKF, Deadband Ablation, and Power

Tables 4B and 5 and Figures 3A and 2D report the live execution of `build/coralnpu/coralnpu_hil_testbench.c` compiled under strict C99 (`cc -std=c99 -O2 -Wall -Wextra -Werror`). Across 1,000 full-horizon steps ($T_{\max} = 90$ d), our `Q16.16` kernel with `Q8.24` slopes and ΣΔ remainder diffusion matches IEEE-754 `FP64` within a maximum relative error of 0.0563% (terminal error ≤ 0.0296%), while the bare-metal C99 mixed `Q16.16/Q8.24` 7-state Huber-EKF tracks the perturbed 8-state plant under AR(1) noise and 5% 6.5σ spikes with 0.94% heteroplasmy RMSE (switching at step 415, 37.35 d, and restoring $W(T_{\max}) = 1.0518, h(T_{\max}) = 0.0222$). Sweeping $N \in \{500, 1{,}000, 2{,}000, 4{,}000, 8{,}000\}$

Table 4: **(A)** 1,000-Step ($\Delta t = 0.09$ d), 120-Trial Closed-Loop Output-Feedback Monte Carlo Benchmark Under ±20% 7-Rate Mismatch, ±8% Initial-State Uncertainty, AR(1) Noise ($\rho = 0.60$), and 5% 6.5σ Spikes (Stage-1 $\kappa(\mathcal{O}_7) = 745.6$), and **(B)** C99 Multi-Step-Size Quantization Deadband Ablation vs. IEEE-754 `FP64`.

A. Closed-Loop Observer ($N_{\mathrm{MC}} = 120$)	**Median J [IQR]**	**$\hat{h}$ RMSE**	**Premature Sw.**	**Mean $h(T_{\max})$**	**Mean $W(T_{\max})$**	**B. C99 Step N (Δt)**	**Scalar `FP32`**	**Naive `Q16`**	**`Q8.24` No-Rem**	**`Q8.24` + ΣΔ Rem**
Oracle True State ($h_{\mathrm{gate}} = 0.080$)	0.909 [0.838, 0.982]	0.00%	0/120 (0.00%)	0.0255	1.0433	500 (0.1800 d)	0.0001%	13.5107%	0.6016%	**0.0543**%
Standard EKF ($\hat{h}_{\mathrm{gate}} = 0.080$)	0.906 [0.851, 0.981]	2.18%	29/120 (24.17%)	0.0274	1.0411	1,000 (0.0900 d)	0.0002%	23.3105%	1.8271%	**0.0563**%
Huber-EKF + Persist ($\hat{h}_{\mathrm{gate}} = 0.080$)	0.913 [0.857, 0.994]	1.95%	17/120 (14.17%)	0.0255	1.0435	2,000 (0.0450 d)	0.0005%	39.6056%	2.8409%	**0.0554**%
Huber-EKF + Margin ($\hat{h}_{\mathrm{gate}} = 0.065$)	**0.933** [0.873, 1.002]	**1.93**%	**6/120 (5.00**%)	**0.0218**	**1.0479**	4,000 (0.0225 d)	0.0010%	84.4063%	22.0179%	**0.0534**%
Stage-1 Gramian $\sigma_1 \dots \sigma_7$:	[963.44, 253.33, 205.73, 171.19, 137.38, 15.61, 1.29] ($\sigma_8 = 0$)					8,000 (0.01125 d)	0.0022%	100.000%	17.4389%	**0.0539**%

Table 5: Full-Horizon (1,000-Step) 8-State `Q16.16` 4-Stage `RK4` Parity Against IEEE-754 `FP64` (`build/coralnpu/coralnpu_hil_testbench.c`) and Saltelli Quasi-Monte Carlo Global Sobol Sensitivity Indices on $\log_{10}(1 + J(\mathbf{u}))$ (2,560 Full ODE Runs).

State Variable	**Terminal `FP64`**	**Terminal `Q16.16` Int**	**Terminal `Q16.16` Float**	**Terminal Err**	**1,000-Step Max / RMS Err**	**Sobol Parameter**	**Viable Range**	S_i	S_{T_i}
x_0: $W(T_{\max})$	1.048706	68,729	1.048721	0.0015%	0.0028% / 0.0015%	$k_{\mathrm{mito},D,\max}$	[0.085, 0.145]	0.739	0.859
x_1: $D(T_{\max})$	0.026781	1,755	0.026779	0.0074%	0.0464% / 0.0137%	k_{clr}	[0.120, 0.240]	0.125	0.125
x_2: $B(T_{\max})$	0.040012	2,623	0.040024	0.0296%	**0.0563**% / 0.0301%	$k_{\mathrm{rep,adv}}$	[1.050, 1.500]	0.102	0.138
x_3: $E(T_{\max})$	0.630235	41,303	0.630234	0.0001%	0.0026% / 0.0011%	θ_{GSH}	[1.020, 1.250]*	0.023	0.023
x_4: $G(T_{\max})$	0.318642	20,883	0.318649	0.0022%	0.0047% / 0.0020%	—	—	—	—
x_5: $N(T_{\max})$	1.469839	96,326	1.469818	0.0014%	0.0063% / 0.0023%	*Lowering $\theta_{\mathrm{GSH}} \in [0.92, 1.02]$ below EP(0)			
x_6: $C(T_{\max})$	1.299926	85,192	1.299927	0.0001%	0.0061% / 0.0030%	crosses the ROS cliff ($S_{T_i}(\theta_{\mathrm{GSH}}) = 0.989$).			
x_7: $M_8(T_{\max})$	1.736594	113,816	1.736694	0.0058%	0.0154% / 0.0085%	C99 Huber-EKF: $\hat{h}$ RMSE 0.94%, $W_{\mathrm{term}} = 1.0518$.			

($\Delta t \in [0.180, 0.01125]$ d) demonstrates why ΣΔ remainder diffusion is essential for integer SIMD: naive round-to-nearest `Q16.16 RK4` diverges from 13.51% error at $N = 500$ to 100.00% deadband stall at $N = 8{,}000$, and `Q8.24` slopes without remainder still degrade to 17.44%–22.02% at $N \geq 4{,}000$, whereas `Q8.24` slopes with 8-lane ΣΔ remainder diffusion maintain 0.0534%–0.0563% error across the entire 16× step-refinement range (compared with 0.0001%–0.0022% for single-lane scalar `RV32F FP32`). Across 2,560 full ODE integrations, Saltelli Sobol indices on $\log_{10}(1 + J(\mathbf{u}))$ confirm that within the viable physiological box ($\theta_{\mathrm{GSH}} \in [1.02, 1.25]$), Stage-1 mitophagy capacity dominates variance ($S_i = 0.739, S_{T_i} = 0.859$), followed by k_{clr} (0.125/0.125) and $k_{\mathrm{rep,adv}}$ (0.102/0.138), whereas dropping θ_{GSH} below EP(0) into $[0.92, 1.02]$ triggers a sharp trans-cliff collapse transition ($S_{T_i}(\theta_{\mathrm{GSH}}) = 0.989$).

Table 6 details both the compiler-verified single-bank L1 DTCM layout (`sizeof(coralnpu_dtcm_bank0_workspace_t)` = 3,696 B out of 8,192 B in `Bank 0`, leaving 4,496 B Bank 0 headroom and 24 KB in `Banks 1-3` power-gated) and the host-emulated cycle-weighted power estimate at 500 MHz (22 nm ULP CMOS model): sequential execution over 3,006 cycles (6.012 μs) yields a cycle-weighted mean active NPU burst power of 3.090 mW (18.58 nJ/burst), accounted separately from external analog front-end sensor excitation (1.800 mW during its 15 ms ADC window).

6 Discussion, Limitations, and Reproducibility Manifest

Limitations and Scope of Claims. First, all ODE trajectories, unconstrained optimal control solutions, mismatched EKF benchmarks, and C99 fixed-point results are computational and host-emulated *in silico* evaluations; no prospective wet-lab, animal, or clinical claims are made. Second, as Figure 2B and Table 4A establish, biological controllability and indirect observability have hard physical limits: initial wild-type depletion inside Zone 0 (18.9% of the (W_0, k_{dmg}) grid) is irreversibly unrescuable without exogenous organelle transplantation, and four-channel proxy sensing leaves pre-switch $\kappa(\mathcal{O}_7) = 745.6$ ill-conditioning ($\sim 1.9\%$ heteroplasmy RMSE under ±20% kinetic mismatch) requiring conservative threshold margin tightening ($\hat{h}_{\mathrm{gate}} = 0.065$). Third, discrete-copy stochastic drift at ultra-low mtDNA copy number, spatially resolved axonal fission-fusion reaction-diffusion PDEs, and gate-level RTL power synthesis on physical FPGA or Coral NPU silicon lie outside the deterministic single-compartment scope of this paper.

MIASE Reproducibility Manifest. Following Minimum Information About a Simulation Experiment (MIASE) guidelines [21], every figure, table entry, and numerical claim in this manuscript is deterministically generated from two zero-stub executable artifacts:

1. **ODE, Adjoint Optimal Control, Mismatched EKF, Phase-Plane & Sobol Suite:** Run `python3 simulator/generate_paper1_verified_benchmarks.py` (fixed PRNG seed `20261109`), which integrates all three non-homothetic 8-state ODE regimes, solves the 5-start 300-variable discrete-adjoint `L-BFGS-B` optimal control problem, computes the Stage-1 7-state Observability Gramian and 1,000-step 120-trial ±20% mismatched closed-loop Huber-EKF benchmark, maps the 31 × 31 (W_0, k_{dmg}) phase plane, runs the 1,000-stream AR(1) persistence and 2,560-run Saltelli Sobol audits, compiles/parses the C99 testbench, writes `simulator/paper1_verified_benchmarks.json`, and renders Figures 1–3.

Table 6: Compiler-Verified Single-Bank L1 DTCM Memory Map (`sizeof(coralnpu_dtcm_bank0_workspace_t)` = 3,696 B in `Bank 0` of 32 KB Coral NPU DTCM) and Cycle-Weighted Host-Emulated Active NPU Power vs. External Sensor Power at 500 MHz (22 nm ULP Process Model).

L1 DTCM Bank 0 Struct Field Group	Bytes	% of 8 KB Bank	Sequential Active NPU Compute Phase	Cycles C_i	Cycle Fraction f_i	Phase Power / Weighted
INT8 Gate (1,216 B) + Goertzel/EMA LUTs (768 B)	1,984 B	24.22%	1. RVV `Zve32x RK4` + Sequential Huber-EKF	1,840	61.21%	3.45 mW → 2.112 mW
Mixed `Q16.16`/`Q8.24` 7-State EKF ($\mathbf{P}, \mathbf{F}, \mathbf{Q}, \hat{\mathbf{x}}$)	452 B	5.52%	2. 16-Bin Goertzel/FFT & 64-Frame Gate	640	21.29%	2.60 mW → 0.554 mW
4-Stage `Q8.24 RK4` (128 B) + $\Sigma\Delta$ (64 B) + Params (132 B)	324 B	3.96%	3. Supervisory Threshold & Rollback Logic	526	17.50%	2.42 mW → 0.424 mW
64-Frame Ring (136 B) + $\mathbf{x}_{\text{ckpt}}$ (32 B) + Stack (768 B)	936 B	11.43%	**Total NPU Compute Burst** (6.012 μs)	**3,006**	**100.00%**	**Mean Active: 3.090 mW**
Verified `sizeof` Active Bank 0 Workspace	**3,696 B**	**45.12%**	Deep-Sleep Retention (Bank 0 Active, Banks 1–3 Gated)	—	> 99.99% duty	18.0 μW continuous
Unallocated Bank 0 Free Headroom	4,496 B	54.88%	External Bio-Impedance/ADC Front-End (15 ms/frame)	—	Separate rail	1.800 mW during window

2. **Bare-Metal C99 `Q16.16 RK4`, Sequential Huber-EKF & Deadband Ablation Testbench:** Compile and run `cc -std=c99 -O2 -Wall -Wextra -Werror build/coralnpu/coralnpu_hil_testbench.c -lm -o build/coralnpu/coralnpu_hil_testbench && ./build/coralnpu/coralnpu_hil_testbench`, verifying 1,000-step `Q16.16` parity (`MaxErr = 0.0563%`), 256-trial 8-lane `Zve32x` widening parity (`MaxULPDiff = 0`), closed-loop C99 Huber-EKF (`h_RMSE = 0.94%`, `sizeof = 3696B/8192B`), and the 5-step-size quantization deadband ablation.

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